DATA REPORT · MARCH 2026

THE STATE OF AI-DRIVEN SEO IN NUMBERS.

We analyzed 342,257 content pieces across 57 brands to find out what actually works in SEO right now — and what doesn't. Performance sections use the last 90 days; structural analysis uses Full available warehouse history (2025-10-13 to 2026-03-25).

flyrank.com

What This Paper Covers

Start here if you want the scope, the questions under review, and the intended reading order without spoiling the findings.

342,257
content pieces analyzed

57
brands included

11
public myth tests

We analyzed 342,257 content pieces across 57 brands. Recent performance uses the last 90 complete days. Structural analysis uses Full available warehouse history (2025-10-13 to 2026-03-25).

The report is built to read in sequence: first the direct portfolio findings, then a single myth-testing block, then the technical appendix, then the operating playbook that leads into the FlyRank close. That order matters because the myth pages reuse definitions introduced in the study setup, and the playbook works best after the evidence is established.

QUESTIONS TESTED IN THIS EDITION

TEST **How much do age, freshness, CTR, engagement, and channel mix explain portfolio performance?**
The findings section establishes the observed patterns before any myth claim is tested.

TEST **Do common SEO beliefs hold once we compare them against the actual portfolio data?**
The myth section keeps every public myth test in one place.

TEST **What should teams do differently after reading the evidence?**
The playbook turns the strongest signals into a refresh and optimization workflow.

MYTH SECTION STRUCTURE

TOTAL	ORDER	FORMAT	REVEAL
11	1 block	1 topic each	In section
Public myth pages	All myth tests grouped together	Same structure across pages	Outcomes withheld until Part III

How To Use This Report

"Read the findings first, then the myth tests. Use the ML appendix for supporting detail, then finish with the playbook."

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The reading order is intentional: findings first, myth tests second, ML appendix third, action pages last.

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Reading Shortcut

If you are short on time, read Executive Summary ’ Core Findings ’ Myth Tests ’ Priority Actions. Use Study Setup for definitions and caveats, and use the ML appendix as supporting detail before the action pages.

About This Study

This paper examines 342,257 content pieces across 57 brands. Recent performance uses the last 90 days: 477,248,928 search impressions,1,537,659 clicks, 1,672,511 sessions, and 17,693 AI sessions. Structural analysis uses Full available warehouse history (2025-10-13 to 2026-03-25).

We asked simple questions: Does updating old content work? Does longer content rank better? Does AI-written content get penalized? Then we let the numbers answer. Direct data comparisons lead the paper. Machine-learning models appear in a bonus appendix.

Every finding is backed by the data. If something didn't hold up, we say so.

Every finding comes with a step-by-step action you can take this week.

342,257

content pieces analyzed across 57 brands

90-DAY IMPRESSIONS	90-DAY CLICKS	SESSIONS TRACKED
477.2M	**1,537,659**	**1.7M**
Last 90 complete days	Organic search	GA4 recent window

AI SESSIONS	AI SHARE	AVG HEALTH SCORE
17,693	**1.06%**	**22.4**
Known AI referrals	Of tracked sessions	Current FlyRank composite context

FULL-HISTORY IMPRESSIONS	FULL-HISTORY CLICKS	FULL-HISTORY SESSIONS
625.8M	**2,126,739**	**2.1M**
Full available warehouse history (2025-10-13 to 2026-03-25)	Warehouse totals	164 active days

30-DAY TREND VS PREVIOUS 30 DAYS

+66.7%	+48.3%	+33.4%
Impressions	Clicks	AI Sessions

Chart read: These show whether things got better or worse in the last month compared to the month before — a quick pulse check, not a substitute for the 90-day comparisons in the main findings.

This dataset includes everything — brand new articles, established pages, and struggling ones. That mix is what makes these patterns real, not cherry-picked. Recent performance uses the 90-day window, while structural analysis uses Full available warehouse history (2025-10-13 to 2026-03-25). Query relevance uses a separate window: Full available query history (2025-06-12 to 2026-03-24).

Study Scope & Evidence Standard.

Here's exactly what we measured and the rules we follow for interpreting the data.

This paper covers 342,257 content pieces across 57 brands, with 477,248,928 impressions, 1,537,659 clicks,1,672,511 sessions, and 17,693 AI sessions.

Reporting windows: Last 90 complete days for current performance, Last 30 complete days for short-term momentum, Full available warehouse history (2025-10-13 to 2026-03-25) for structural analysis, and Full available query history (2025-06-12 to 2026-03-24) for ranking-query relevance and query-coverage analysis.

VERIFIED TOTAL	VALUE
Content	342,257
Brands	57
Impressions	477,248,928
Clicks	1,537,659
Sessions	1,672,511
AI Sessions	17,693

Metric Windows

Last 90 complete days for current performance, Last 30 complete days for short-term momentum, Full available warehouse history (2025-10-13 to 2026-03-25) for structural analysis, and Full available query history (2025-06-12 to 2026-03-24) for ranking-query relevance and query-coverage analysis.

Data Source

This paper uses local recent-snapshot exports plus locally materialized full-history page and query exports. Each section is labeled with the window that matches the question being answered.

Evidence Standard

Headline findings prioritize direct aggregate comparisons. ML pages are exploratory appendix material and do not override direct portfolio evidence.

Interpretation Rule

External SEO beliefs are treated as hypotheses, not proof. When direct portfolio evidence conflicts with industry narratives, the portfolio result wins.

Understanding the Metrics

This paper leads with real search performance numbers. We use our composite score for context, not as the main proof.

Health Score (0-100)

Impressions (30 pts) + position (30 pts) + CTR (20 pts) + scroll depth (20 pts). Higher = better performing content. This is a FlyRank metric, so major findings also show raw search numbers alongside it.

Position Tiers

Top 3: positions 1-3 (the best spots) | Page 1: positions 4-10 | Striking Distance: positions 11-20 (close to page 1 — highest ROI zone) | Page 3-5: positions 21-50 | Deep: 50+ (very hard to find)

Age & Freshness

Age = how old the article is. Freshness = how recently it was updated. These are different things: a 2-year-old article updated last week is old but fresh. A 3-month-old article never touched since launch is young but stale.

AI Traffic

Visits from AI tools like ChatGPT, Perplexity, Gemini, Copilot, Claude, and Meta AI. Currently 1.1% of tracked sessions (17,693 of 1.7M). Small but growing.

Optimization Flags

Labels FlyRank assigns to pages that need attention — like 'Fix CTR' (people see it but don't click) or 'Zombie Page' (basically invisible). They tell you what to fix, not that the content is bad.

Trend Direction

Based on the last 30 days vs the 30 days before. Up: >10% growth. Down: >10% decline. Stable: within +/-10%. Flat: not enough data. New: published within 30 days.

Finding Tags

CONFIRMED Hypothesis validated by data

REVERSED Data showed the opposite

NUANCED Partially true with caveats

ML Techniques Used

We also ran machine learning models (clustering, prediction, and pattern detection) as a bonus appendix. Most published comparisons use large buckets, and smaller cells are called out where they appear. These models explore the data — the main findings rely on direct number comparisons.

Methodology & Limitations.

Data Sources
Data comes from Google Search Console, Google Analytics 4, and revenue data — all stored in BigQuery with read-only access (we can look but never change the source data).

Study Scope
Last 90 complete days for current performance, Last 30 complete days for short-term momentum, Full available warehouse history (2025-10-13 to 2026-03-25) for structural analysis, and Full available query history (2025-06-12 to 2026-03-24) for ranking-query relevance and query-coverage analysis.

Evidence Standard
Headline findings prioritize direct aggregate comparisons. ML pages are exploratory appendix material and do not override direct portfolio evidence.

Health Score
Impressions (30 pts) + position (30 pts) + CTR (20 pts) + scroll depth (20 pts).

ML Pipeline
212.4K active pages (with real sessions and impressions). Models: K-Means clustering (5 groups), Random Forest, Logistic Regression, PCA, and Decision Tree — all tested on held-out data (80/20 split). Most published comparisons use large buckets, and smaller cells are explicitly called out as directional when they remain.

Statistical Approach
Correlations use Pearson r. Most main buckets are sized to avoid fragile one-off reads, and small cross-cells are flagged in the narrative where they remain. Percentiles use APPROX_QUANTILES. CTR is weighted (total clicks ÷ total impressions). We do not report p-values or confidence intervals.

Coverage Windows
Full available warehouse history (2025-10-13 to 2026-03-25) for structural sections, Last 90 complete days for current-performance sections, Last 30 complete days for momentum, and Full available query history (2025-06-12 to 2026-03-24) for ranking-query relevance.

Confounding Variables
Content age confounds model comparisons. Revenue tracking covers 39 of 57 clients, so revenue claims are treated cautiously. AI referral detection may miss some sources, and the query-history window is shorter than the page-history window.

Limitations

This is an observational study — we found patterns, not proof of cause and effect. Health score is our own metric, not a Google standard. Query data uses a shorter window than page data. Engagement time comes from aggregated fields, not individual session replay.

Why FlyRank Handles This For You

Most teams can't connect Google Search Console, Analytics, content data, AI traffic, and revenue into one picture. FlyRank does this automatically — keeping measurements connected, surfacing refresh and optimization queues, tracking cannibalization, and turning patterns into repeatable weekly actions.

The Anatomy of Growing Content.

What do growing pages have in common? And what's different about the ones losing traffic?

We split every page into two groups: gaining traffic vs losing traffic. The clearest difference is age. Growing pages are younger and already a little more visible in search. Length is almost the same.

1487
avg words (growing)

1481
avg words (declining)

185d
avg age (growing)

228d
avg age (declining)

DIRECTION	COUNT	WORDS	AGE	AVG IMP	AVG POS	HEALTH
up	74.5K	1.5K	185d	2.9K	16	36.6
down	46.2K	1.5K	228d	2.6K	15.9	32.7

Growing pages average 1.5K words vs 1.5K for declining ones. That word-count gap is basically nothing. But they are **18.9% younger** (185 vs 228 days). Declining pages aren't necessarily bad — many still get traffic. They're just older and losing momentum. With 74.5K rising vs 46.2K falling pages, the age gap is a strong pattern in this dataset.

What to Do With This

Improve pages that already earn impressions

Why: The biggest gap here is age, not length, so start with pages Google already sees and keep them useful

How: List pages with real impressions that still feel thin or incomplete, then add missing subtopics, examples, definitions, and comparison sections instead of padding.

Expected: Gives already-visible pages a better chance to keep growing instead of fading out.

Measure: Track impressions, clicks, and average position 30 and 60 days after the update.

Review aging pages before they drift into decline

Why: The falling cohort is older on average and more likely to have missed a recent refresh cycle

How: Create a quarterly review for pages that are around 6-9 months old, then prioritize the ones that once performed well but have started to flatten.

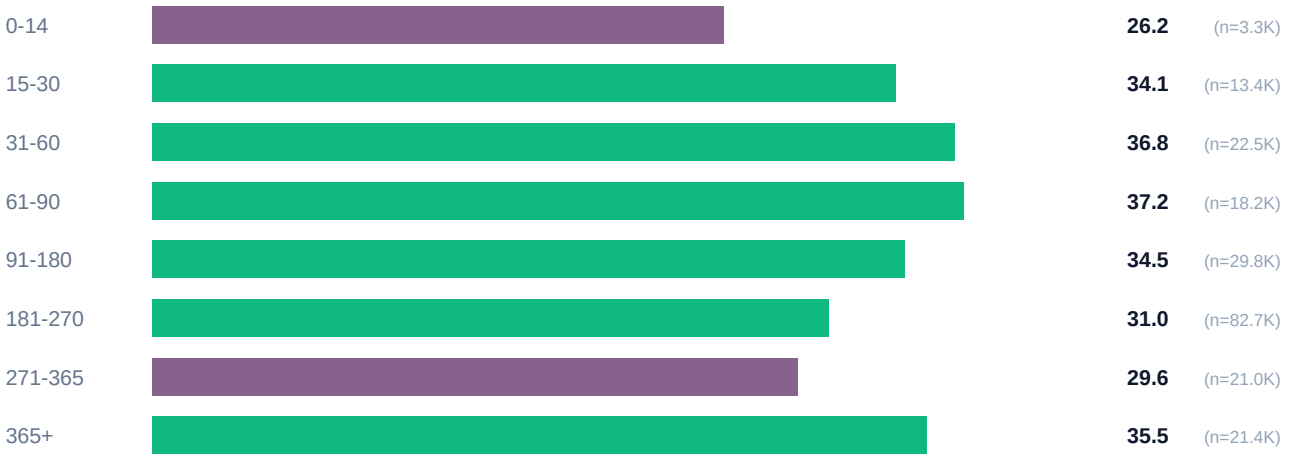
Expected: Protects pages with existing visibility before the decline becomes expensive to reverse.

Measure: Track refresh coverage and 30-day impression change for reviewed versus unrevised pages.

The Content Performance Curve.

Every piece of content has a life cycle. It peaks around 61-90 days, then gets weaker after about 9 months if nobody keeps it sharp.

HEALTH SCORE BY CONTENT AGE (DAYS)



It takes about 60-90 days for Google to fully discover and rank a new page — that's when **performance peaks at 37.2 health**.

After that, performance gradually declines. By 271-365 days, health drops to 29.6 as the content gets older. **But old does not mean dead:** the 365+ group still averages 35.5 health. That tells us age alone does not kill a page. Strong older pages can still hold up, especially when teams keep them maintained.

Create a review cycle before pages hit the 9-12 month mark

Why: Performance starts declining after 270 days — catch pages before the drop

How: Schedule editorial reviews for pages approaching that age, then refresh the ones that already have traffic or strategic value before they go stale.

Expected: Preserves the visibility older pages have already earned.

Measure: Track refreshed-page coverage and compare impression retention before and after the review window.

The Freshness Multiplier.

Content updated 1-3 months ago grows the fastest. Here's how big the freshness effect really is.

GROWTH-TO-DECLINE RATIO BY FRESHNESS WINDOW

Pages updated 0-30 days ago are too new to read reliably. But at **31-90 days, freshness is the strongest measured growth signal** — 5.43:1 growth-to-decline ratio. Pages untouched for 181-360 days show clearly stale behavior at 3.02:1, while the 361+ bucket spikes to 37.19:1 — but that's a tiny sample with only 21 declining page, so don't read too much into it.

The biggest surprise: among pages older than a year, the cohort refreshed in the last 30 days jumped from 23 to 37 health (**1.6x boost**) compared with otherwise old pages last updated 181-360 days ago, and from 82 to 4.2K impressions (**52x**). Refreshing proven content is one of the strongest measured levers in this dataset.

1.6x
health boost from refresh

52x
impression boost

5.43:1
growth ratio at 31-90d

Chart read: Read 31-90 as the strongest stable freshness signal. The 361+ bar is present because it exists in the local sample, but its 37.19:1 ratio is unstable: 781 growing pages versus only 21 declining.

Run a recurring refresh program for mature pages

Why: Refreshing mature pages produces 1.6x health and 52x impressions in this dataset

How: Review older pages with proven historical visibility, refresh outdated facts and examples, add missing subtopics, and resubmit the strongest updates for recrawl.

Expected: Lifts mature pages from roughly 23 health to 37 and materially improves impression potential.

Measure: Track impressions, clicks, and average position on refreshed mature pages versus comparable untouched pages.

The Age-Freshness Matrix.

How old is the content and how recently was it updated? This matrix shows how those two factors combine.

HEALTH SCORE HEATMAP: AGE TIER × FRESHNESS TIER

	0-30	31-90	91-180	181-360	361+
0-14	26.2				
15-30	34.1				
31-90	39.0	35.8			
91-180	36.9	26.7	35.0		
181-365	35.9	16.5	28.9	27.6	
365+	37.0			23.1	32.5

Four key zones:

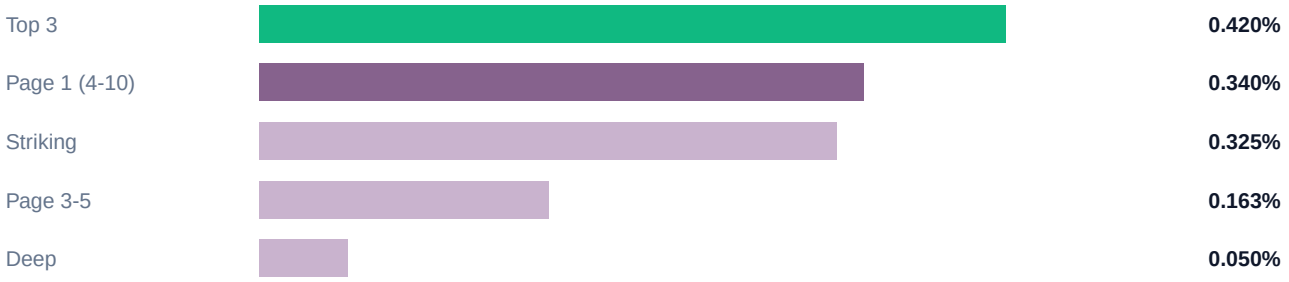
1. **Growth Engine** (31-90d age, recently updated): 39.02 health.
2. **Refresh Winners** (365+, updated within 30d): 36.98 health.
3. **Decay Zone** (181-365d, untouched 6-12 months): 27.57 health.
4. **Survivors** (365+, 361+ stale): 32.48 health — very small sample, so read it cautiously.

The biggest takeaway from this study: a 1-year-old article that you update can compete with brand-new content (36.98 vs 39.02 health). You do not always need net-new content — you often need updated content on proven pages.

Click Capture by Position Tier.

The higher you rank, the more clicks you get — and the drop-off is steep.

WEIGHTED CTR BY POSITION TIER



0.420%
weighted CTR in top 3

0.340%
weighted CTR on page 1

88%
drop from top 3 to deep

This chart shows what percentage of people click on your content based on where it ranks. These are real click-through rates from this portfolio (total clicks ÷ total impressions per tier), not generic benchmarks.

The takeaway: improving a page already on page 1 gets you more clicks faster than trying to rescue a page buried on page 5. Once content drops off the first page, click capture shrinks fast.

Refine search snippets on pages that already rank on page 1

Why: Small ranking and snippet gains on already-visible pages usually convert into click lift faster than rebuilding weak pages from scratch

How: Review pages ranking on page 1, tighten titles and descriptions around clear intent, and test more specific promise statements instead of generic headings.

Expected: Improves click capture on visibility you already own.

Measure: Track weighted CTR, clicks, and average position for the revised group over the next 30 days.

Support positions 11-20 with targeted relevance improvements

Why: Pages just outside page 1 already show search demand and topical fit

How: Refresh the page, improve internal links from stronger related pages, and tighten the opening sections so the page answers the target query faster.

Expected: Increases the chance of moving a borderline page into the higher-click part of the curve.

Measure: Track movement into page 1 and the resulting click lift.

Engagement and Visibility Move Together.

High scroll + high engagement = +16.1 health points. Visibility consistency compounds the effect.

How These Buckets Work

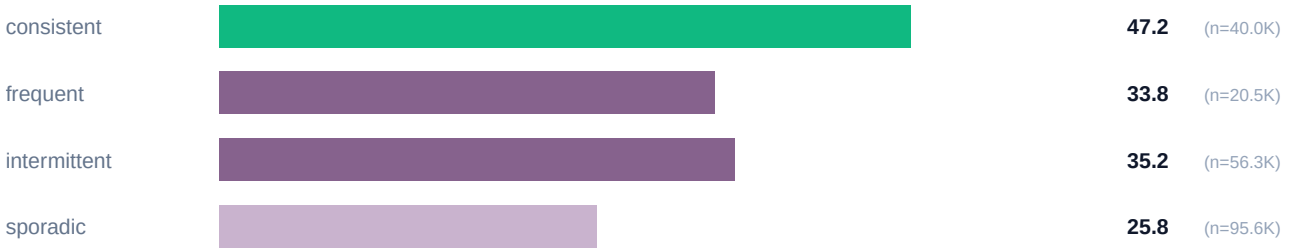
Scroll depth buckets use the share of the page consumed: Low <30%, Mid 30-59%, High 60%+. Engagement-rate buckets use Low <45%, Mid 45-69%, High 70%+. Visibility here means how many of the last 90 complete days the page recorded at least one search impression.

HEALTH SCORE BY SCROLL DEPTH × ENGAGEMENT RATE

	Low (<45%)	Mid (45-69%)	High (70%+)
Low (<30%)	32.0	36.5	26.0
Mid (30-59%)	41.6	45.1	50.0
High (60%+)	40.3	46.3	48.1

Chart read: Darker cells = higher health score. The top-right corner is the target state, but 3 cells have fewer than 50 pages, so treat those corners as directional rather than definitive.

HEALTH BY VISIBILITY CONSISTENCY (DAYS VISIBLE IN 90)



Pages where people scroll deeper and stay engaged score 48.1 health. Pages people bounce from score 32. Consistency matters too: pages visible 80+ days score 47.2, while sporadic pages score 25.8. **That 21.4-point gap is why we use visibility as context in this paper — it means sustained discoverability in search, not just existence in the warehouse.**

Improve scanability on pages that already earn consistent visibility

Why: High-engagement pages score 16.1 health points higher than low-engagement ones

How: Use clearer sections, jump links, tighter intros, and visuals that make the page easier to scan and understand.

Expected: Supports the engagement patterns associated with stronger visibility.

Measure: Track scroll depth, clicks, and average position changes on the revised group.

The Winning Combinations.

The easiest win: target topics where people are ready to buy and where competitors haven't shown up yet.

AVERAGE HEALTH BY INTENT × COMPETITION LEVEL

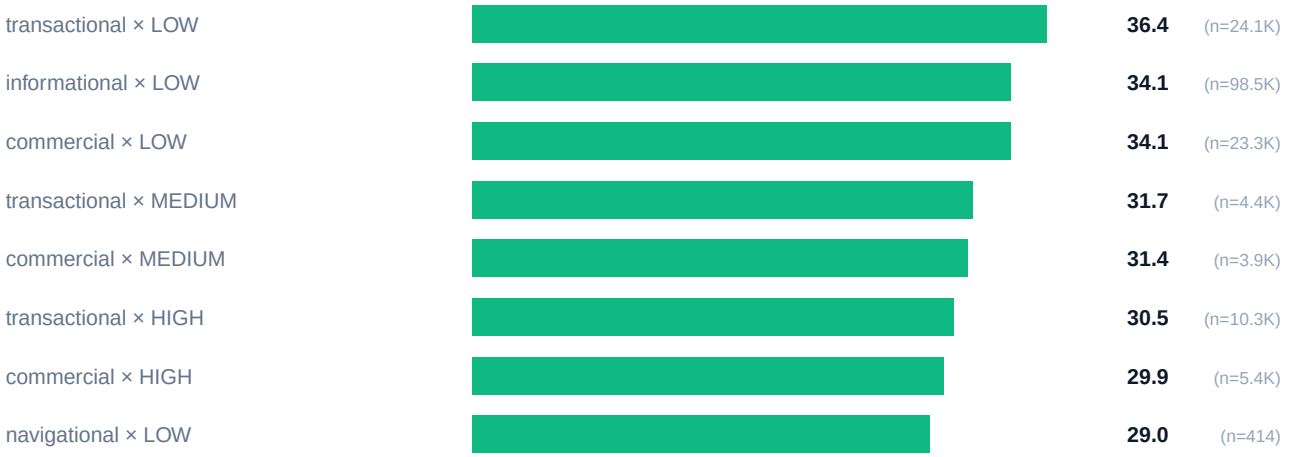


Chart read: Each bar is one intent + competition pairing. The chart is sorted by average health and only includes pairings with at least 100 pages, so tiny edge cases do not drive the ranking.

PAGE-1 CONTENT BY WORD COUNT TIER

Chart read: This only looks at pages already on page 1. It shows how much depth winning pages tend to need — not a rule to make every page longer.

Transactional intent + low competition leads every metric in this portfolio: 36.4 health and 3.8K impressions. Informational + low competition pages average 34.1 health — still solid, but the buy-ready topics clearly outperform.

Among page-1 content, the strongest word-count bucket is **3500+ words** (44.8 health,2.4K impressions). But longer content only wins when the topic genuinely needs that depth — padding doesn't help.

Chart read: Use this page as a prioritization lens, not as a universal rule. Smaller subsegments can still swing around if the sample gets thin, which is why the main ranking view excludes pairings below 100 pages.

Reserve a larger share of the calendar for lower-competition commercial or transactional topics

Why: Those topics outperform the portfolio average in both visibility and FlyRank health context

How: Map topics where the reader is already comparing options, then target the versions where the current result pages are beatable with a stronger page rather than a bigger budget.

Expected: Improves the odds of reaching page 1 faster and capturing more useful demand.

Measure: Track impressions, clicks, and average position by intent class over the next publishing cycle.

Captured Traffic Value.

The defensible proxy here is `clicks × CPC`, not `impressions × CPC`. Window: Full available warehouse history (2025-10-13 to 2026-03-25).

CLICK-EQUIVALENT VALUE BY INTENT

INTENT	PAGES	CLICKS	CPC	VALUE
informational	195.0K	1.1M	$2.79	$480.2K
commercial	45.3K	386.7K	$2.19	$261.3K
transactional	50.5K	576.1K	$2.29	$355.7K
navigational	1.6K	2.5K	$7.62	$3.1K

$1.1M
captured click-equivalent value

$389.9M
impressions × CPC upper bound

0.28%
captured vs upper bound

How much is your organic traffic worth? We multiply **actual clicks × what those clicks would cost in Google Ads** (CPC). This gives you a dollar value for your content without inflating the number with impressions that didn't lead to clicks.

The click-based value is 0.28% of the impressions × CPC number — that gap is exactly why you should never use the inflated version as a headline.

In Full available warehouse history (2025-10-13 to 2026-03-25), **informational** leads total click-equivalent value by scale, while **transactional** has the strongest capture rate against the impressions × CPC upper bound. Value density tells a clearer story than raw traffic volume alone, which is why the upper-bound number is shown only as a caveat.

Use click-equivalent value to rank optimization candidates

Why: It preserves CPC economics without pretending every impression is worth a click

How: Prioritize visible pages where clicks already exist and the underlying CPC is meaningful, especially in transactional and commercial intent buckets.

Expected: Improves prioritization quality when traffic counts alone hide value density.

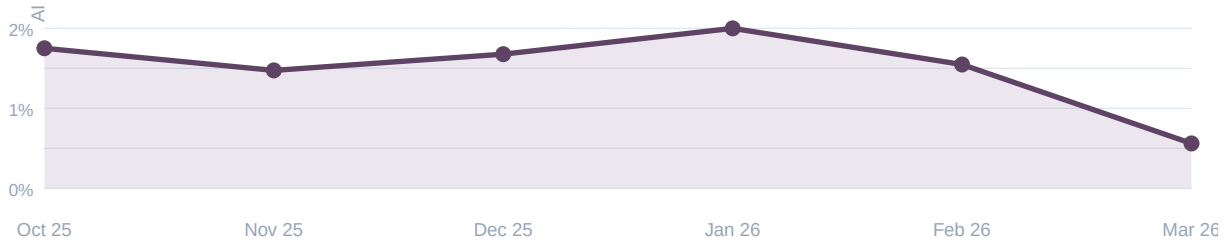
Measure: Track click lift, weighted CPC, and click-equivalent value before and after optimization.

AI Traffic: A Different Signal.

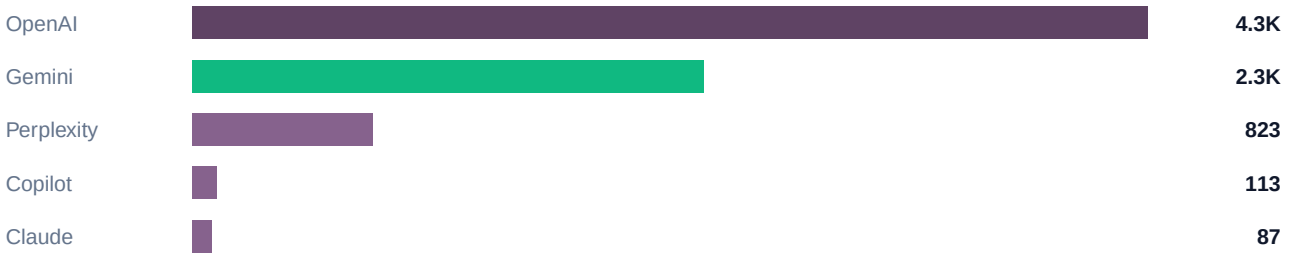
AI traffic does not line up perfectly with normal Google traffic. Here is the pattern we can actually measure.

AI BUCKET	COUNT	HEALTH	AVG IMP	AVG POS
high_ai	2.8K	38.5	5.2K	14
some_ai	2.4K	48.9	20.1K	18.1
no_ai	207.1K	32.8	2.0K	14.3

AI TRAFFIC SHARE (%) — MONTHLY



AI REFERRAL PROVIDERS (LOCAL LAST-30D BREAKDOWN)



1.06%
AI share of portfolio sessions

1.57%
AI page rate in full export

1.9x
OpenAI vs Gemini (30d)

Pages with AI referrals do not look exactly like the pages that win normal search. In this dataset, the high-AI group averages 3x more impressions but slightly weaker Google positions than the no-AI group. So AI visibility and Google visibility overlap, but they are not the same thing.

Reality check: AI traffic is still only 1.06% of tracked sessions (17.7K of 1.7M). It's growing, but it's not replacing Google yet. In this export, the highest AI referral rates show up in commercial intent and the 5K+ length bucket.

Monitor AI referrals separately — they're a different channel

Why: AI traffic is still small, but the pages that get it do not look exactly like normal search winners

How: Review high-impression pages, make key facts easy to scan, and track AI referrals separately from search clicks.

Expected: Prepares your content for AI visibility without pretending the channel is bigger than it is.

Measure: Track AI session share, provider mix, and page-level AI referral rates alongside standard search metrics.

AI Model Performance.

Neither OpenAI nor Gemini content is universally better. When we compare content of the same age, each model wins in different categories.

AGE-CONTROLLED COHORT HEALTH: OPENAI VS GEMINI

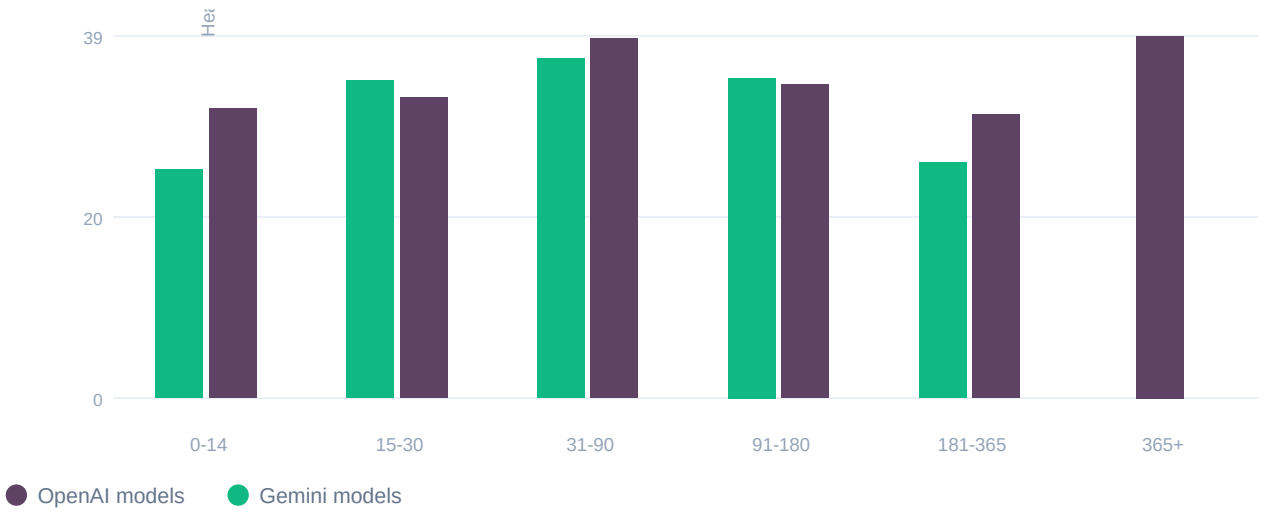


Chart read: Each age tier compares provider families inside the same publication band. Read it as a cohort comparison, not as proof that one provider family wins everywhere.

PROVIDER	COUNT	HEALTH	AVG IMP	AVG POS
OpenAI	73.7K	32.54	1.4K	13.9
Gemini	72.1K	33.51	2.1K	11.5

Once we control for age, Gemini leads some cohorts and OpenAI leads others — **the AI model matters less than how well you edit and publish the content.**

Output quality, editing standards, topic fit, and rollout timing are the real differentiators. This dataset does not support a blanket penalty tied only to AI usage.

Compare content systems within the same age and topic cohorts

Why: The model picture changes once age mix is controlled

How: When you test writing systems or prompts, review them inside the same publication window and topic class rather than comparing a newer batch to an older installed base.

Expected: Produces cleaner decisions about which production workflow actually performs better.

Measure: Track impressions, average position, and health context within matched cohorts at 30, 60, and 90 days.

Portfolio-Level Trends.

How portfolio metrics evolved over Full available warehouse history (2025-10-13 to 2026-03-25).

MONTHLY IMPRESSIONS

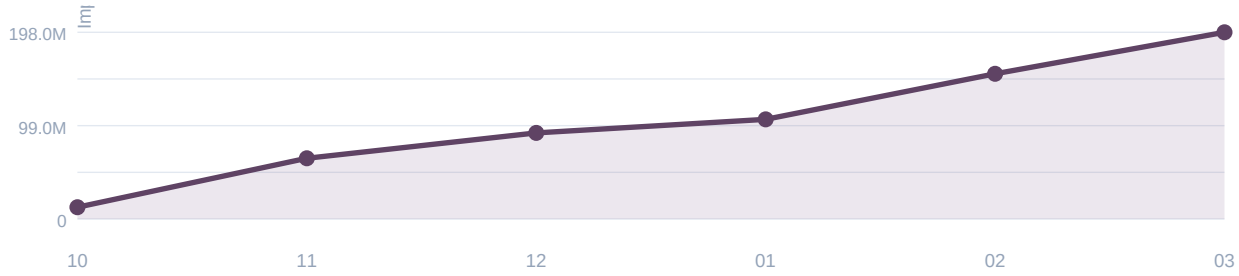


Chart read: Read this as a portfolio trend line, not a page-level benchmark. A rising line means the overall content base is earning more visibility over time, while a flat or falling line signals slowing momentum.

MONTH	IMPRESSIONS	CLICKS	SESSIONS	AI SESSIONS	ACTIVE CONTENT
2025-10	12.5M	75.6K	22.2K	422	17.3K
2025-11	64.4M	271.7K	229.9K	3.7K	93.5K
2025-12	91.3M	297.2K	208.2K	3.8K	98.4K
2026-01	105.7M	375.4K	219.4K	4.8K	110.0K
2026-02	154.0M	508.7K	345.3K	5.8K	156.4K
2026-03	198.0M	598.1K	1.1M	6.6K	180.2K

This is the big picture — every month of data we have (Full available warehouse history (2025-10-13 to 2026-03-25)). Use it to see whether the portfolio is growing overall and whether recent trends match the longer pattern.

Across this window, impressions, clicks, sessions, and active content all trend upward. AI sessions are growing too, but they're still a much smaller layer than classic organic traffic.

Keyword Drift.

You pick a keyword, publish content for it, then Google decides what the page actually ranks for. Here's how often that matches.

51.5%
on-target (25%+ from keyword)

48.5%
drifted (<25% from keyword)

11.5%
zero matching queries

HOW MUCH TRAFFIC COMES FROM YOUR TARGET KEYWORD

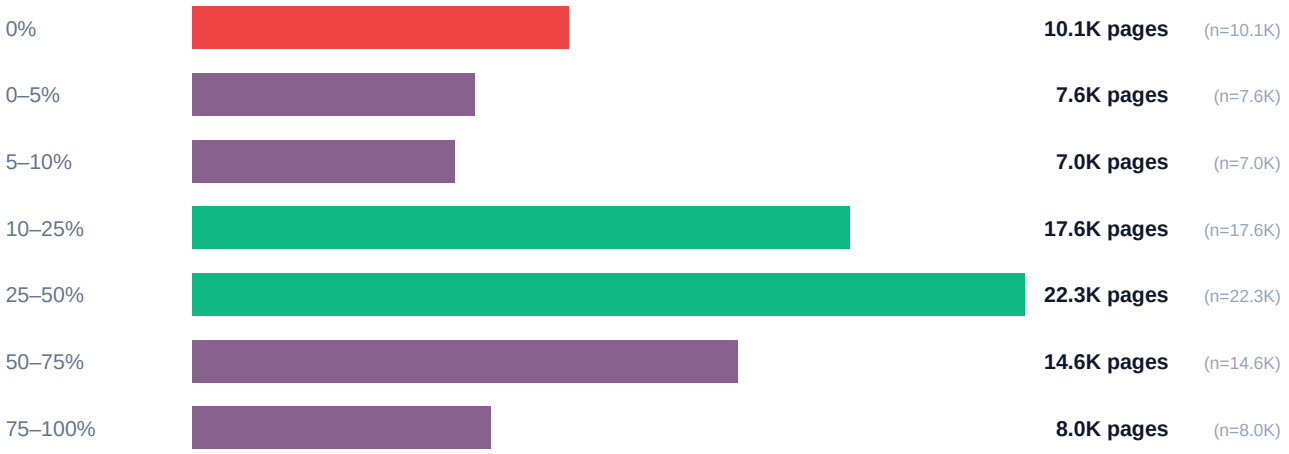


Chart read: Each bar is a group of pages. The label shows what percentage of that page's total impressions come from queries matching its primary keyword.

On-Target Pages

AVG IMPRESSION SHARE
53.6%
AVG QUERIES PER PAGE
125.6
MATCHING QUERIES
18.5

Drifted Pages

AVG IMPRESSION SHARE
8.8%
AVG QUERIES PER PAGE
185
MATCHING QUERIES
9.3

We compared the keyword each page was created for (the primary keyword in FlyRank) against every query that page actually ranks for in Google Search Console. A query "matches" if it's the same phrase or shares 50%+ of the same words.

About half the portfolio (51.5%) gets 25% or more of its traffic from the target keyword. The other half drifted — Google sends them traffic for different queries. And 11.5% of pages get zero impressions from any query matching their keyword.

Drifted pages aren't failing — they rank for 185 queries on average (more than on-target pages at 125.6). They found an audience, just not through the keyword you planned. The median page gets 25.9% of its traffic from its target keyword. The top 10% get 73%+.

How We Measured This

For each of 87.1K pages with both a primary keyword and Google Search Console query data, we summed all impressions from queries that match the keyword (exact string match or 50%+ word overlap), then divided by the page's total impressions. Pages where this share is 25% or higher are "on-target." Below 25% means the page drifted to different queries. Window: Full available query history (2025-06-12 to 2026-03-24).

Drift vs Performance.

Does matching your target keyword predict better performance? We checked impressions, clicks, CTR, and health score.

AVERAGE IMPRESSIONS BY KEYWORD MATCH SHARE

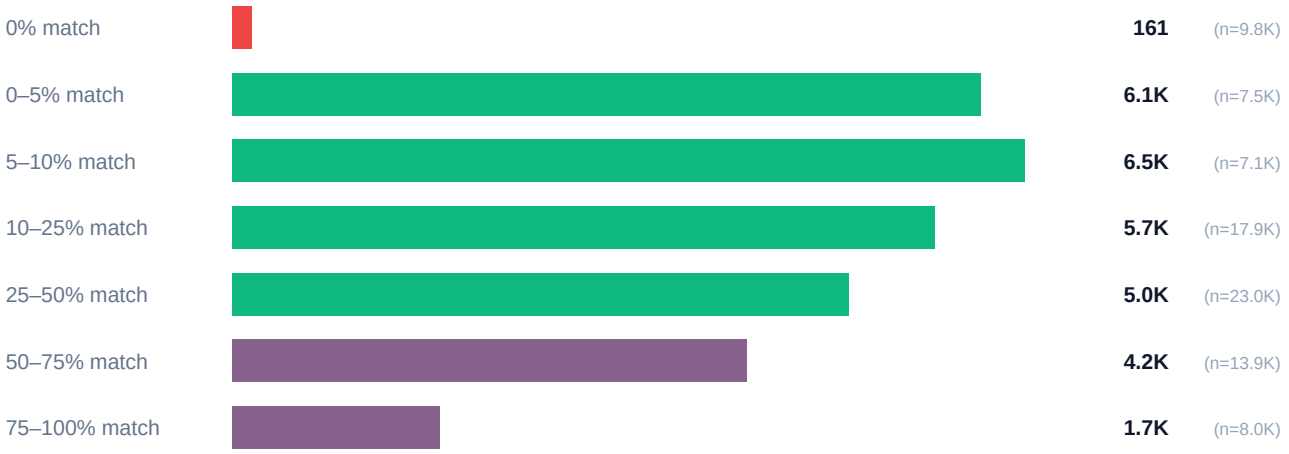


Chart read: This shows average impressions per page, grouped by how much of the page's traffic comes from its target keyword. Higher keyword match does not mean more traffic.

AVERAGE HEALTH SCORE BY KEYWORD MATCH SHARE

5–10%
highest impressions bucket

75–100%
highest CTR bucket

160.5
avg impressions at 0% match

Keyword match share does not predict impressions, clicks, or health score. The Pearson correlations are near zero: impressions -0.039, clicks -0.006, health -0.013. Only CTR shows a slight positive trend (0.033).

The bucket breakdown tells the real story. Pages with **0% keyword match average only 160.5 impressions** — these pages have almost no traction at all. But the sweet spot is 5–10% match at **6.5K avg impressions and 40.3 health**. Pages tightly locked onto one keyword (75-100% match) average only 0 impressions.

Drift is not a problem — it's a feature. The best-performing pages capture their target keyword plus a wide net of related queries. Pages too tightly matched to one keyword have less total reach. The only real concern is the 0% group — pages with no keyword traction at all.

Don't chase 100% keyword match — aim for broad relevance

Why: Pages with 5–10% keyword match outperform tightly-matched pages by 6.5K impressions on average

How: Write content that covers the full topic around your keyword, not just the exact phrase. Include related questions, comparisons, and subtopics. Let Google match you to the queries your content genuinely answers.

Expected: Broader content earns more impressions while still capturing the target keyword as one of many traffic sources.

Measure: Track total impressions and query count per page. A healthy page should rank for dozens of queries, not just the one you targeted.

What Weakened, Reversed, or Stayed Nuanced.

Not everything we expected to work actually did. Here are the SEO beliefs this edition challenges, in the same order used in the detailed myth pages.

11

published myth tests

3

reversed myths

7

nuanced myths

1

direct debunk

These are popular SEO beliefs that our data proved wrong or incomplete. We could have hidden the messy parts to tell a cleaner story, but knowing what weakens or breaks is just as valuable as knowing what confirms.

REVERSED

High search volume is not a reliable traffic forecast

Monthly search volume behaves more like a competition signal than a page-level traffic ceiling in this portfolio.

REVERSED

Optimization flags mean measurable opportunity, not automatic failure

Flagged pages are often the visible pages worth improving. Read flags as workflow priority, not as a claim that more flags are inherently better.

NUANCED

Word count helps only when it buys better coverage

Longer pages can win more demand when they answer more questions. Extra words without relevance do not create performance by themselves.

DEBUNKED

AI-generated content is not penalized by default

This dataset does not show a blanket penalty for AI use. Editing quality, topic fit, and publishing process matter more.

NUANCED

Keyword difficulty matters, but mostly as a growth environment signal

Lower-competition topics are safer, but difficulty does not work as a simple yes-or-no predictor by itself.

NUANCED

Freshness amplifies quality rather than replacing it

Refreshing strong pages works. Refreshing weak pages helps less. The best observed health cell is not always the freshest one.

NUANCED

Publishing more content helps — but a strong library beats a busy one

High-velocity publishers average better health, but the single healthiest brand publishes nothing. Volume without quality is noise.

REVERSED

Higher CPC keywords deliver worse organic performance

Every step up in CPC predicts worse health, fewer impressions, and worse position. The cheapest keywords outperform across every organic metric.

NUANCED

Engagement metrics reflect quality but do not drive rankings

High engagement correlates with health but shows near-zero correlation with position. Position drives engagement opportunity, not the reverse.

NUANCED

No single search intent type is a performance silver bullet

Transactional intent leads slightly in health, but informational content grows faster and dominates volume. Unclassified content is the real underperformer.

NUANCED

Consistent visibility predicts health — but not growth

The most visible content has plateaued. The real growth engine is moderate consistency, where content has proven viability but still has room to climb.

"High Search Volume = More Traffic."

Search-volume numbers from SEO tools are monthly estimates. When you compare them on the same monthly basis, they still fail as reliable traffic forecasts.

Why The Monthly Comparison Matters

DataForSEO search volume is a monthly estimate. That makes 90-day page impressions divided by three the clean apples-to-apples benchmark. We still show the raw 90-day comparison and the longer full-history check for context.

SHARE OF PAGES BEATING MONTHLY SEARCH VOLUME

Chart read: This uses 90-day impressions divided by three so page traffic is compared against a monthly keyword-volume benchmark on the same basis. The 10K+ bucket is real but thin (330 pages), so treat the top end directionally.

SV BUCKET	COUNT	AVG SV	90D IMP	90D / 3	BEAT SV
1-100	96.5K	23	2.5K	837	65%
100-1K	16.2K	306	2.9K	982	34%
1K-10K	3.8K	2.8K	2.2K	748	7%
10K+	330	30.3K	3.0K	997	1%

58.59%
monthly-adjusted pages beating volume

68.2%
pages beating volume (90-day)

50.32%
pages beating volume (full history)

The clean comparison is monthly: **58.59% of pages still beat stored search volume after dividing 90-day impressions by three**. In the raw 90-day view, 68.2% beat it. Over the longer full-history window, 50.32% still beat it.

The correlation between search volume and actual traffic is effectively zero: 0.0003 for recent data and -0.0016 over the longer window. Low-volume keywords are the easiest to outperform, while high-volume terms are the least likely to

beat their benchmark. **Search volume tells you more about competition than about the traffic a page will actually earn.**

Use search volume as a rough guide, not a traffic prediction

Why: Actual page traffic only weakly correlates with keyword volume numbers

How: Start with beatable demand and clear intent match. Treat search volume as a competition signal, not an expected traffic ceiling.

Expected: Helps you find topics where real upside is higher than keyword tools suggest.

Measure: Track average position, page-level impressions, and the share of pages exceeding their keyword-volume benchmark.

"Content With Flags Is Failing."

Pages with diagnosed issues aren't failing — they're visible enough to measure. That makes them your best optimization targets.

About These Flags

These are FlyRank's internal optimization flags: diagnostic labels we assign to pages with enough visibility to measure. They are not external quality signals from Google.

HEALTH SCORE BY NUMBER OF ACTIVE OPTIMIZATION FLAGS

Chart read: Read this as workflow signal, not as a reward for accumulating flags. The pattern is not monotonic: 0 flags = 33.1, 2 flags = 28, 3 flags = 45.1, and 5 flags is a tiny sample (81 pages).

This was one of the most counterintuitive findings. In this dataset, pages with diagnosed issues are often the pages visible enough for us to measure in the first place. Pages with **3+ flags materially outperform** the zero-flag cohort, but the curve is not perfectly linear, so do not read this as "more flags is better."

Why? A page needs to be visible enough for us to even detect problems. No flags often means nobody's looking at your page — including Google. Pages with multiple diagnosed issues have real traffic that can be improved.

A page with a specific measurable problem is often more valuable than a page nobody sees. Flags = opportunity, not failure. Read this as workflow priority, not as a public claim that a higher flag count is inherently desirable.

Treat measurable optimization issues as priority opportunities

Why: A page with a specific diagnosed issue is usually easier to improve than a page with no measurable demand

How: Review visible pages for the exact problem they show, such as weak click capture or thin coverage, then fix that issue first instead of rewriting the whole asset.

Expected: Focuses effort on pages where the upside is knowable and more immediate.

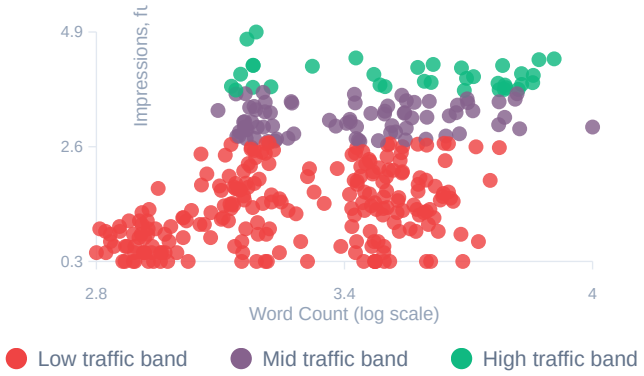
Measure: Track impressions, clicks, and average position before and after each targeted fix.

"Longer Content Always Ranks Better."

Longer content does get more traffic — but because it covers more topics, not because Google rewards word count. Page data uses Full available warehouse history (2025-10-13 to 2026-03-25); query data uses Full available query history (2025-06-12 to 2026-03-24).

AVERAGE IMPRESSIONS BY WORD COUNT BUCKET

IMPRESSIONS VS WORD COUNT



CLICKS VS CHARACTER COUNT

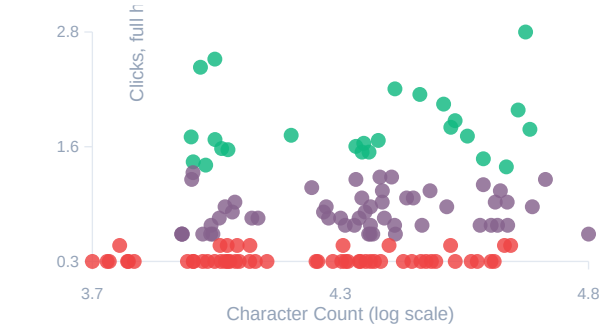


Chart read: These are descriptive scatter plots, not fitted trend lines. The spread is broad rather than tightly linear, which is exactly the point: longer pages and larger character counts do not produce traffic in a simple straight-line way.

WORDS	PAGES	AVG IMP	AVG SEC	QUERIES	OFF-TARGET %	TOP REL.
<1K	230.4K	1.6K	0.49	170.5	69.4	6.2
1K-1.5K	50.7K	690	1.41	63.58	73.35	4.84
1.5K-2K	39.0K	932	1.62	74.06	69.33	5.66
2K-2.5K	8.0K	433	1.81	41.32	76.76	4.22
2.5K-3.5K	59.0K	625	1.67	66.35	66.61	6.91
3.5K-5K	28.6K	1.2K	1.82	79.17	65.75	7.4
5K+	11.5K	9.8K	3.13	466.86	80.46	6.47

5K+
Most queries answered

0.4127
Word count vs queries link

-0.0565
Word count vs time-on-page link

The 5K+ bucket gets the most traffic and covers the most queries (5K+). But **longer content wins because it answers more questions, not because it's long.**

We do not see a strong relationship between word count and engagement time (correlation: -0.0565). Length doesn't equal engagement. Only 6.17% of pages have their top query as an exact keyword match, while 60.97% are led by an off-target query — which is why depth needs a relevance check, not just more words. **Write as much as the topic needs — no more.**

Use depth deliberately on pages that already show demand

Why: Longer buckets in the combined page-history and query-history export carry more impressions and broader query coverage

How: Expand thin but visible pages with missing subtopics, examples, comparisons, and evidence instead of padding every page to a fixed word target.

Expected: Improves visibility where additional depth is most likely to matter without encouraging irrelevant query drift.

Measure: Track impressions, query count, and off-target impression share by refreshed word-count bucket.

"AI-Generated Content Is Penalized."

Almost all content in this dataset is AI-generated. This dataset does not show a blanket penalty tied only to AI use.

Does Google penalize AI-generated content? Our dataset of 342.3K pieces — almost all AI-generated across multiple model cohorts — gives us a clear view.

When we compared models within the same age tiers, both stronger and weaker outcomes showed up inside AI-authored cohorts. No single model always wins. **The differences come from how well the content was edited and published, not from whether AI wrote it.**

Model choice, editing standards, and topic fit drive the differences we see. This dataset does not support a blanket penalty narrative tied only to AI usage.

Evaluate publishing workflows by results, not assumptions

Why: The differences in this portfolio come from editing and process quality, not a blanket AI penalty

How: Test writing systems within the same topic and publication window, then compare actual performance instead of assuming one method is safer.

Expected: Keeps content decisions evidence-led and avoids false binary thinking about AI use.

Measure: Track impressions, average position, and quality-review outcomes by workflow cohort.

"Keyword Difficulty Is a Reliable Predictor."

Competition level matters — but not in the way you'd expect. The growth ratio tells the real story.

LOW Competition

HEALTH	GROWTH RATIO
34.4	2.1:1
AVG POSITION	COUNT
14.5	146.4K

HIGH Competition

HEALTH	GROWTH RATIO
29.6	1.2:1
AVG POSITION	COUNT
16.5	19.9K

GROWTH-TO-DECLINE RATIO BY COMPETITION LEVEL

Chart read: This compares the odds of pages growing versus declining. Longer bars mean a safer growth environment, not a guarantee that every keyword will be easy.

LOW competition pages score 34.4 health vs HIGH at 29.6 — a 4.8-point gap. But **the growth ratio is where the real difference shows**: LOW competition has a 2.1:1 growth-to-decline ratio vs 1.2:1 for HIGH.

That means LOW competition keywords are **75% more likely to be growing** than HIGH competition ones. So competition still matters, but it does not tell the whole story by itself. It is more useful as a growth difficulty signal than as a simple yes-or-no ranking rule.

"Fresh Content Always Outperforms."

Updating a great page makes it amazing. Updating a weak page makes it slightly less weak. Here's the data.

HEALTH SCORE BY FRESHNESS TIER × WORD COUNT TIER

	<1000	1000-2000	2000-3500	3500+
0-30	19.9	35.5	33.9	38.7
31-90	17.9	17.7	34.7	29.0
91-180	32.3	30.1	32.0	39.1
181-360	23.9	20.0	25.5	25.4

Chart read: Darker cells = stronger health. Read this as the interaction between freshness and depth, not as a rule that every page should be newer or longer.

At 3500+ words: fresh content (0-30d) scores 38.7 health vs stale (181-360d) at 25.4 — a **13.3-point gap**. At <1000 words, that relationship does not hold in the same way: fresh scores 19.9 vs stale at 23.9, a 4.0-point deficit for the freshly updated thin-page cohort.

Freshness multiplies existing quality. Refreshing a strong, comprehensive page produces dramatic results. Thin pages do not show the same payoff here, which is exactly why freshness should be read together with depth and coverage. Also note the top cell in this matrix is not the freshest one: **91-180 × 3500+** reaches **39.1**. Prioritize refreshing your best content first, but do not collapse this into a simplistic "newest always wins" story.

"Publishing More Content = Better Results."

We compared all 58 brands in the portfolio by how much they published in the last 90 days vs their actual performance.

PERFORMANCE BY PUBLISHING VELOCITY (CONTENT-WEIGHTED)

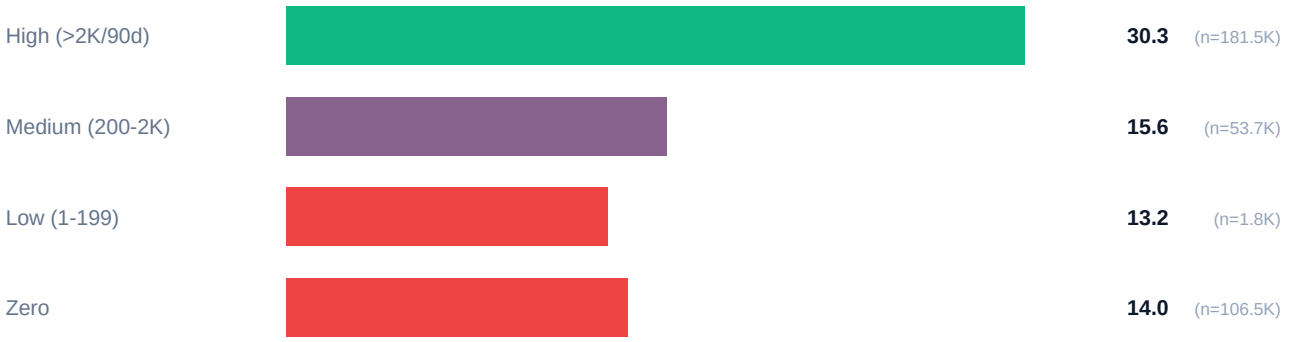


Chart read: Health is weighted by content volume — larger libraries pull the average harder. Count shows total content pieces in each velocity tier.

45.4
best non-publisher (outdoor subscription box)

45.3
best high-velocity (speech therapy app)

15.14:1
growth ratio (best active)

High-velocity publishers (> 2,000 pieces in 90 days) average **30.3 health** — roughly double the zero-publishing group (14). That looks decisive, but the exceptions tell a richer story.

A brand in the **outdoor subscription box niche reached 45.4 health and 57.0M impressions with zero recent publishing**. That is the single highest health score in the entire portfolio. Meanwhile, a speech therapy app brand published nearly 3.0K pieces in 90 days and reached 45.3 — almost identical health, plus a 15.14:1 growth ratio.

Publishing velocity helps on average, but a strong dormant library can outperform an active mediocre one. The real lesson: volume is a multiplier of quality, not a replacement for it.

Match publishing speed to content quality — never sacrifice quality for volume

Why: High-velocity publishers average 2x the health of non-publishers, but the portfolio's healthiest brand publishes nothing

How: Set a velocity target that your team can sustain without dropping quality. Review health scores monthly — if health drops while volume rises, slow down.

Expected: Avoids the trap of high-volume, low-quality publishing that dilutes portfolio health.

Measure: Track average health score per publishing cohort monthly. Watch for velocity-health divergence.

"Higher CPC Keywords = Better Organic."

SEO practitioners often chase high-CPC keywords assuming they signal high organic value. Our data says the opposite.

HEALTH SCORE BY CPC BUCKET (343K CONTENT)

Chart read: The pattern is perfectly monotonic and inverse. Every step up in CPC predicts worse organic health.

CPC BUCKET	COUNT	HEALTH	AVG IMP	AVG POS	CTR	UP/DOWN
$Under 0.50	23.1K	27.2	2.3K	17.7	22.0%	2.19
$0.50-0.99	8.4K	25.8	1.6K	18.8	24.2%	1.92
$1.00-1.99	11.0K	25	1.6K	18.9	24.2%	1.68
$2.00-4.99	13.5K	24.3	1.4K	19.7	22.2%	1.08
$5.00+	12.2K	19.9	1.2K	25.1	29.1%	1.28

-0.045
CPC ” health correlation

-0.017
CPC ” impressions correlation

0.102
CPC ” position correlation

The cheapest keywords (<$0.50) reach **27.2 health and 2.3K avg impressions**, while the most expensive ($5+) manage just **19.9 health and 1.2K impressions** — at position 25.1.

All three correlations are effectively zero or slightly negative. **CPC measures ad-market competition, not organic opportunity.** Targeting expensive keywords is a paid-search strategy that does not translate into organic performance.

Stop using CPC as a proxy for organic keyword value

Why: CPC correlates at -0.045 with health and -0.017 with impressions — effectively zero

How: When evaluating keywords for organic content, prioritize search volume, competition level, intent match, and topic relevance over CPC. Reserve CPC analysis for paid campaigns.

Expected: Redirects effort toward keywords where organic upside is real rather than where ad markets are expensive.

Measure: Compare organic impressions and health for new content targeted at low-CPC vs high-CPC keywords.

"Better Engagement = Higher Rankings."

Google has repeatedly denied that dwell time and bounce rate are direct ranking factors. Does our data agree?

Important Context

293.2K of 343.6K content pieces (85%) have no engagement or scroll data at all. The patterns below reflect the subset that does have data, which skews toward higher-performing content.

HEALTH SCORE: ENGAGEMENT × SCROLL MATRIX

0.015	0.056	0.115	0.087
engagement ” position	scroll ” position	engagement ” health	scroll ” health

High engagement + high scroll content reaches **48.3 health at position 14**. Content with no engagement data sits at **19.4 health at position 16.4**. That is only a 2.4 position gap between the best and worst engagement cohorts.

The correlation between engagement and position is **0.015 — essentially zero**. Scroll rate and position: 0.056. Both are negligible. Engagement does correlate weakly with health (0.115), but health includes engagement in its formula, so that is partly circular.

Position drives engagement opportunity, not the other way around. Higher-ranked pages get more visitors who can engage. This supports Google's public statements: engagement metrics are not direct ranking signals.

Optimize engagement for user value, not for rankings

Why: Engagement ” position correlation is 0.015 — near zero

How: Improve scroll depth, time on page, and interaction for conversion and reader satisfaction rather than treating them as ranking levers. Focus ranking efforts on relevance, authority, and technical SEO.

Expected: Avoids wasted effort on engagement hacks that do not move organic position.

Measure: Track engagement and position independently. If engagement improves but position does not, the data confirms the pattern.

"Transactional Intent Always Wins."

SEO teams often prioritize transactional keywords for their assumed conversion value. Does intent type actually predict organic performance?

HEALTH SCORE BY SEARCH INTENT (343.6K CONTENT)

INTENT	COUNT	HEALTH	AVG IMP	POS	CTR	UP/DOWN
Informational	183.6K	22.6	1.5K	16.9	33.8%	1.86
Transactional	50.1K	28.8	2.4K	15.4	27.4%	1.71
Commercial	43.7K	27.1	2.2K	17.5	27.9%	1.83
Navigational	1.3K	15.8	585	25.4	33.6%	1.42

53.4%
portfolio is informational

28.8
transactional health (best)

1.86:1
informational growth ratio (best)

Transactional intent leads in health (28.8) and impressions (2.4K), but only by a small margin over commercial (27.1). The gap between transactional and informational is just 6.2 health points.

Informational content has the best growth ratio (1.86:1) — more content is trending up vs down compared to transactional (1.71:1). Informational also holds the highest CTR (33.8%) and dominates volume at 53.4% of the portfolio.

Notably, unclassified content (64.8K pieces) has a 0.56:1 growth ratio — declining almost 2x faster than growing. Intent classification itself correlates with performance.

Balance intent mix rather than over-indexing on transactional

Why: Informational content grows at 1.86:1 vs transactional at 1.71:1 — faster growth at scale

How: Maintain a healthy mix of informational and transactional content. Use informational pieces to build topical authority and funnel visitors toward transactional pages.

Expected: Captures the full demand spectrum rather than competing only on high-intent, high-competition terms.

Measure: Track growth ratio, health, and impressions by intent bucket monthly.

"Consistent Visibility = Consistent Growth."

If your content appears in search results every day, it must be growing — right? This is the most counterintuitive finding in our study.

HEALTH SCORE VS GROWTH RATIO BY VISIBILITY CONSISTENCY

Chart read: Bar length = health score. Color = growth ratio: green (>2:1 = growing), amber (1-2:1 = mixed), red (<1:1 = declining). The most visible content is healthiest but NOT the fastest growing.

47.7
health (very consistent)

3.59:1
growth ratio (moderate = best)

0.37:1
growth ratio (rare = worst)

146.1-K
zero-visibility content

The "Very Consistent" group (80-90 days visible) has the **best health (47.7) and most impressions (8.3K)** — but the **worst growth ratio (1.09:1)**. Nearly equal up and down. This content has peaked.

The real growth engine is the **Moderate group (30-59 days): 3.59:1 growth ratio** — by far the best in the dataset. This content has proven viability but still has room to climb.

Below 10 days of visibility, content is in decline territory: **0.37:1** — declining 2.7x faster than growing. And 146.1K content pieces (42.5% of the portfolio) have zero visibility at all.

The strongest correlation in the entire study is visibility consistency ” health at**0.686** (strong). But visibility predicts where you are, not where you are going. The growth sweet spot is moderate consistency — enough to prove viability, not so much that you have already plateaued.

Focus optimization energy on moderately visible content (30-59 days)

Why: This tier has a 3.59:1 growth ratio — 3x better than the most consistent tier

How: Identify content visible 30-59 days in the last 90. These are your highest-leverage optimization targets: proven demand, room to climb.

Expected: Concentrates effort where the growth probability is highest.

Measure: Segment content by visibility days. Track how many pages move from moderate to consistent tier after optimization.

Structural Risks.

Portfolio-wide issues that hold back performance even when individual pages are good.

723.4K
cannibalizing queries

76.7M
impressions at risk

292.8K
clicks at risk

Critical 66.7 % High 8.7 % Medium 14.9 % Low 9.8

Chart read: Red = critical overlap between your own pages. The more red, the more urgently you need to consolidate or clarify which page should rank for each topic.

INDEXING COVERAGE

confirmed indexed 91.4 % never indexed

confirmed indexed (194.1K) never indexed (18.2K)

Chart read: Green = pages Google can find. Red/yellow = pages that are invisible to search. Fixing indexing issues can unlock traffic before you create any new content.

What Actually Correlates?

Which metrics move together? This chart shows how 16 different measurements relate to each other across 211.1K active pages.

FEATURE CORRELATION HEATMAP (PEARSON R)

	health score	impressions	clicks	sessions	ai sessions	scroll rate	engagement rate	ctr	avg position	content age_days	days since_update	word count	search volume	cpc	competition	days with_impressions
health score	1.0	0.3	0.2	0.2	0.1	0.2	0.1	0.1	-0.5	-0.1	-0.2	0.1	-0.0	-0.1	-0.1	0.5
impressions	0.3	1.0	0.7	0.4	0.2	-0.0	0.0	-0.0	-0.1	0.1	-0.1	0.0	0.0	-0.0	-0.0	0.3
clicks	0.2	0.7	1.0	0.4	0.1	0.0	0.0	0.0	-0.1	0.0	-0.1	0.0	-0.0	-0.0	-0.0	0.2
sessions	0.2	0.4	0.4	1.0	0.3	0.0	0.1	0.0	0.0	-0.0	-0.1	0.3	-0.0	-0.0	-0.0	0.2
ai sessions	0.1	0.2	0.1	0.3	1.0	0.0	0.0	-0.0	0.0	0.0	-0.0	0.1	-0.0	-0.0	-0.0	0.1
scroll rate	0.2	-0.0	0.0	0.0	0.0	1.0	0.3	0.0	-0.1	-0.0	-0.0	0.2	0.0	-0.0	-0.1	-0.0
engagement rate	0.1	0.0	0.0	0.1	0.0	0.3	1.0	0.0	-0.0	0.0	-0.1	0.1	0.0	-0.0	-0.0	0.1
ctr	0.1	-0.0	0.0	0.0	-0.0	0.0	0.0	1.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.1
avg position	-0.5	-0.1	-0.1	0.0	0.0	-0.1	-0.0	-0.0	1.0	0.1	-0.1	-0.1	0.0	0.1	0.1	0.0
content age_days	-0.1	0.1	0.0	-0.0	0.0	-0.0	0.0	0.0	0.1	1.0	0.3	-0.6	0.1	0.1	0.1	0.3
days since_update	-0.2	-0.1	-0.1	-0.1	-0.0	-0.0	-0.1	0.0	-0.1	0.3	1.0	-0.3	-0.0	-0.0	-0.1	-0.4
word count	0.1	0.0	0.0	0.3	0.1	0.2	0.1	-0.0	-0.1	-0.6	-0.3	1.0	-0.0	-0.1	-0.1	-0.1
search volume	-0.0	0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	0.0	0.1	-0.0	-0.0	1.0	0.0	0.0	0.0
cpc	-0.1	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	0.1	0.1	-0.0	-0.1	0.0	1.0	0.3	0.0
competition	-0.1	-0.0	-0.0	-0.0	-0.0	-0.1	-0.0	-0.0	0.1	0.1	-0.1	-0.1	0.0	0.3	1.0	0.0
days with_impressions	0.5	0.3	0.2	0.2	0.1	-0.0	0.1	-0.1	0.0	0.3	-0.4	-0.1	0.0	0.0	0.0	1.0

Chart read: Dark cells = these metrics rise and fall together. If two metrics are highly correlated, improving one likely means you've already improved the other — don't count it as two separate wins.

STRONGEST POSITIVE CORRELATIONS

Impressions × Clicks: **r=0.728**
health score × Days Visible: **r=0.487**
Clicks × Sessions: **r=0.429**
Impressions × Sessions: **r=0.4**
Sessions × ai Sessions: **r=0.346**

STRONGEST NEGATIVE CORRELATIONS

Content Age × Word Count: **r=-0.552**
health score × Average Position: **r=-0.457**
Days Since Update × Days Visible: **r=-0.362**
Days Since Update × Word Count: **r=-0.266**
health score × Days Since Update: **r=-0.166**

Chart read: These are the strongest pairings from the heatmap. They show which metrics tend to move together — not proof that changing one will automatically change the other.

The Content Archetypes.

We grouped all active content into 5 types using machine learning. These are patterns the algorithm found — not rigid categories.

AVERAGE HEALTH SCORE BY CONTENT ARCHETYPE

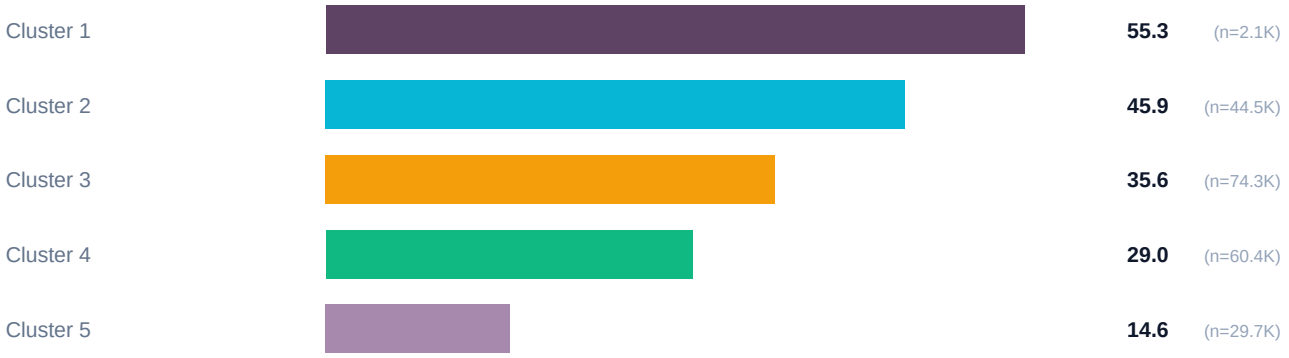


Chart read: This is the fastest way to compare the cluster groups. Higher bars mean stronger average health; the count tells you how much of the portfolio sits in that archetype.

Cluster 1

(1% of content, n=2.1K)

Exploratory segment with 1% of the sampled ML set (n=2124).

HEALTH	IMPRESSIONS	AVG POSITION	AGE	WORDS
55.33	53.2K	13.8	210d	3.9K

These are your top performers. Protect them, keep facts fresh, and do scheduled maintenance before they slip.

Cluster 2

(21.1% of content, n=44.5K)

Exploratory segment with 21.1% of the sampled ML set (n=44535).

HEALTH	IMPRESSIONS	AVG POSITION	AGE	WORDS
45.86	4.9K	10.1	275d	473

These are your top performers. Protect them, keep facts fresh, and do scheduled maintenance before they slip.

Cluster 3

(35.2% of content, n=74.3K)

Exploratory segment with 35.2% of the sampled ML set (n=74309).

HEALTH	IMPRESSIONS	AVG POSITION	AGE	WORDS
35.57	1.4K	10.6	82d	3.4K

These are middle-of-the-pack pages. Better clarity, stronger coverage, and fresher details can push them into the top tier.

Cluster 4

(28.6% of content, n=60.4K)

Exploratory segment with 28.6% of the sampled ML set (n=60371).

HEALTH	IMPRESSIONS	AVG POSITION	AGE	WORDS
29.01	**531**	**7.2**	**232d**	**736**

These are middle-of-the-pack pages. Better clarity, stronger coverage, and fresher details can push them into the top tier.

Cluster 5

(14.1% of content, n=29.7K)

Exploratory segment with 14.1% of the sampled ML set (n=29735).

HEALTH	IMPRESSIONS	AVG POSITION	AGE	WORDS
14.58	**200**	**45.3**	**244d**	**707**

These are underperformers. Rewrite, consolidate, or retarget them instead of assuming a light refresh will be enough.

Chart read: Each card is a recurring page pattern in the portfolio. Use the interpretation line as the action cue, not as a hard label you must preserve forever.

These clusters help you see common content states in the portfolio. Compare the profiles to understand what different types of content look like — but use the main findings, not these groups, for decisions.

What Predicts Health?

Which factors matter most for FlyRank health score? We trained a model to predict the score and checked which inputs it relied on most. This explains model behavior, not independent causality.

About Visibility Inputs

Days Visible means the number of the last 90 complete days where the page recorded at least one search impression. It is not page age, and it is not just how long the page has existed in BigQuery.

FEATURE IMPORTANCE (RANDOM FOREST ' HEALTH SCORE)

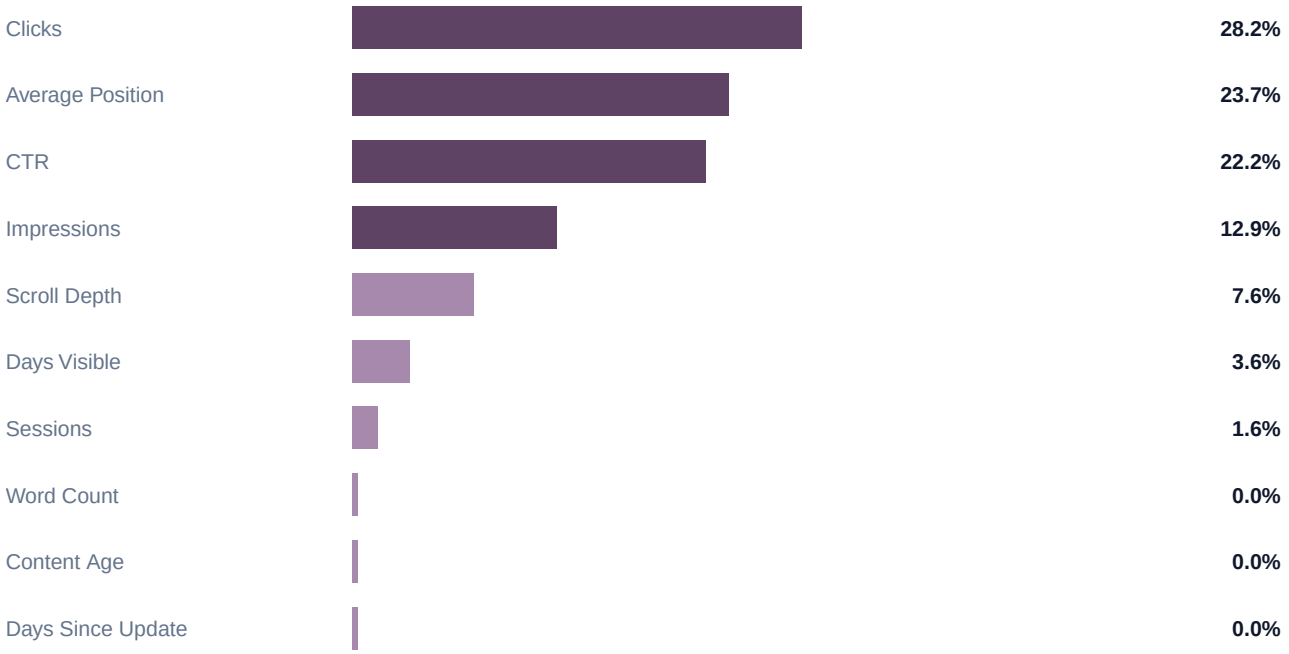
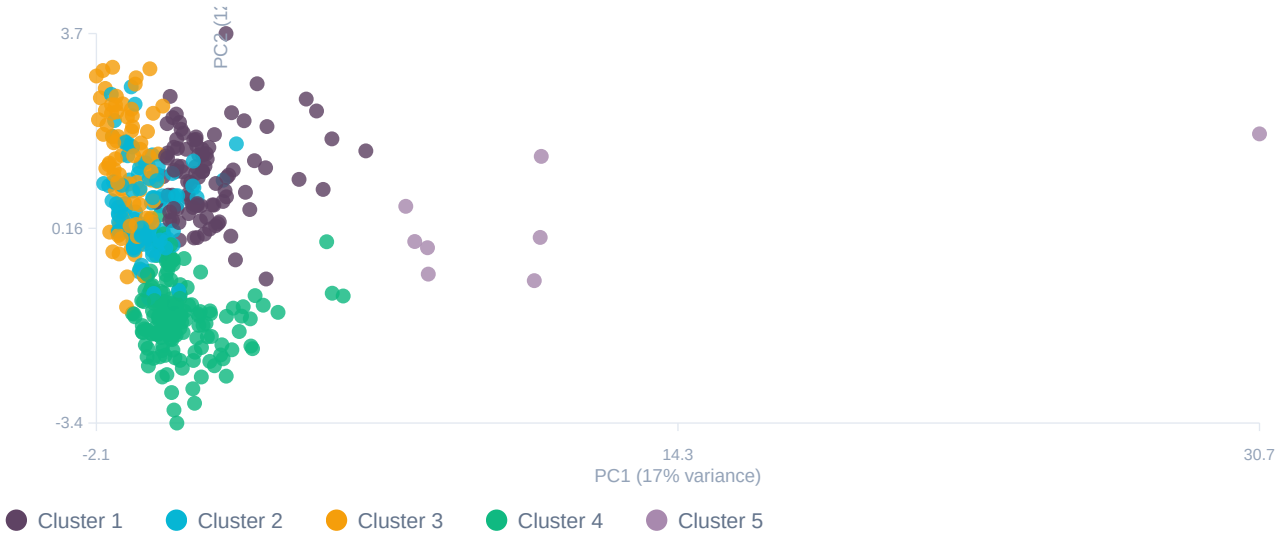


Chart read: Taller bars = the model relied more on that factor. Since health score already includes visibility inputs, this shows what the model uses, not necessarily what you should optimize first.

Clicks is the #1 predictor at 28% importance, followed by Average Position (24%) and CTR (22%). Note: health score is partly built from visibility inputs, so high importance for those features is expected — this shows what the model relies on, not a guaranteed optimization order.

PCA 2D PROJECTION — COLORED BY CLUSTER



How to Read This

Each dot is one page compressed into 2 dimensions. Tight color groups mean similar content. Overlap between colors means the cluster boundaries are fuzzy, not cleanly separated.

Chart read: Each dot is one page, compressed into 2D. Tight color groups = similar content. Overlap between colors = fuzzy boundaries between content types.

What Predicts Growth?

Logistic regression (68% holdout accuracy) describing which sampled features separate growing from declining pages.

GROWTH PREDICTION COEFFICIENTS (LOGISTIC REGRESSION)

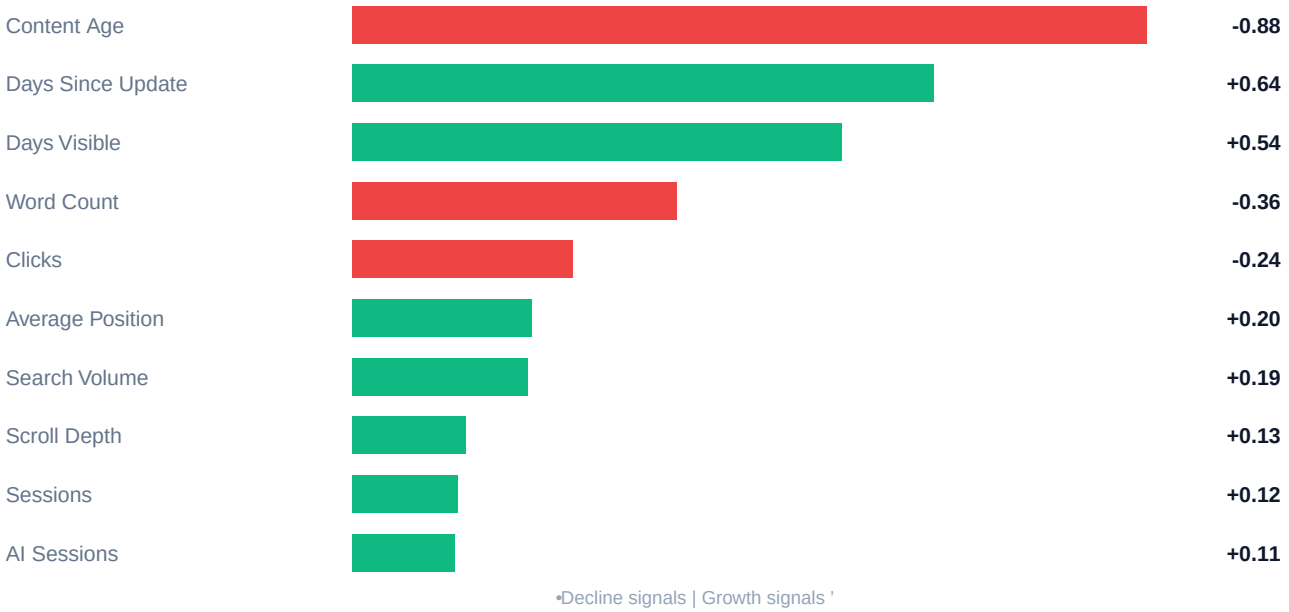


Chart read: Content Age is the biggest drag on growth. Days Since Update and Days Visible push pages toward growth. Use this to know where to look, not as a guaranteed formula.

The #1 factor dragging pages down: age. The #1 factors pushing pages up: being consistently visible and getting recent impressions. Older, forgotten pages decline. Active, visible pages grow.

The decision tree (also holdout-tested) shows which combinations of visibility, clicks, and engagement separate stronger pages from weaker ones — useful for pattern recognition, not as a guaranteed formula.

TOP 10% VS BOTTOM 10% CONTENT PROFILE

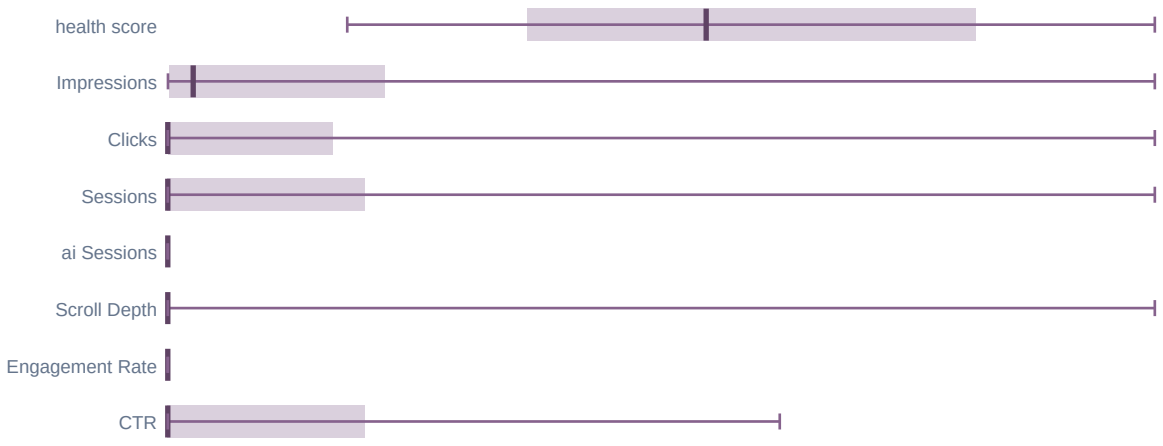
DECILE	HEALTH	AVG IMP	POS	AGE	FRESH	WORDS	COUNT
top_10pct	60.7	12.4K	5	195d	66d	1.5K	21.2K
bottom_10pct	10	46	32	214d	78d	1.3K	21.2K

Chart read: Top row = what the best 10% of pages look like. Bottom row = the worst 10%. If your page looks like the bottom row, you know what needs to change.
The top 10% of pages tend to be fresher, longer, and much better positioned than the bottom 10%. Use this as a benchmark to compare against — not as a guaranteed recipe.

Metric Distributions.

Where does your content stack up on the core performance outputs? This page shows the spread for the outcome metrics most readers care about first.

PERCENTILE BOX PLOTS (P10 / P25 / P50 / P75 / P90)



Whisker = P10-P90 Box = P25-P75 Median = P50

Interpretation Guide

These charts do not mean the portfolio sits in the top 10%. Each row is normalized independently, so the visual shows distribution shape within that metric. Use the table for the real values.

Chart read: Each row is scaled independently so you can compare shape, not absolute size. Use the table below for the actual values and the median line to see what a typical page looks like for that specific metric.

METRIC	P10	P25	MEDIAN	P75	P90	MEAN
health score	10	20	30	45	55	33.11
Impressions	1	8	129	1103	5013	2235.41
Clicks	0	0	0	2	12	7.25
Sessions	0	0	0	2	10	7.92
ai Sessions	0	0	0	0	0	0.08
Scroll Depth	0	0	0	0	37.5	9.82
Engagement Rate	0	0	0	0	0	1.2
CTR	0	0	0	0.2	0.62	0.49

What To Do With Percentiles

Below P25 means the page is in the bottom quarter for that specific metric. P50 is the typical page. Above P75 means the page is unusually strong on that metric relative to the portfolio, not that the whole portfolio is top tier.

Use these percentiles to see where a page falls relative to the rest of the portfolio. The chart is normalized for readability, so always use the table for real values.

Input & Context Distributions.

These are the structural and context metrics behind the outcomes. They are useful for calibration, but they should not be read as direct performance proof.

PERCENTILE BOX PLOTS (P10 / P25 / P50 / P75 / P90)

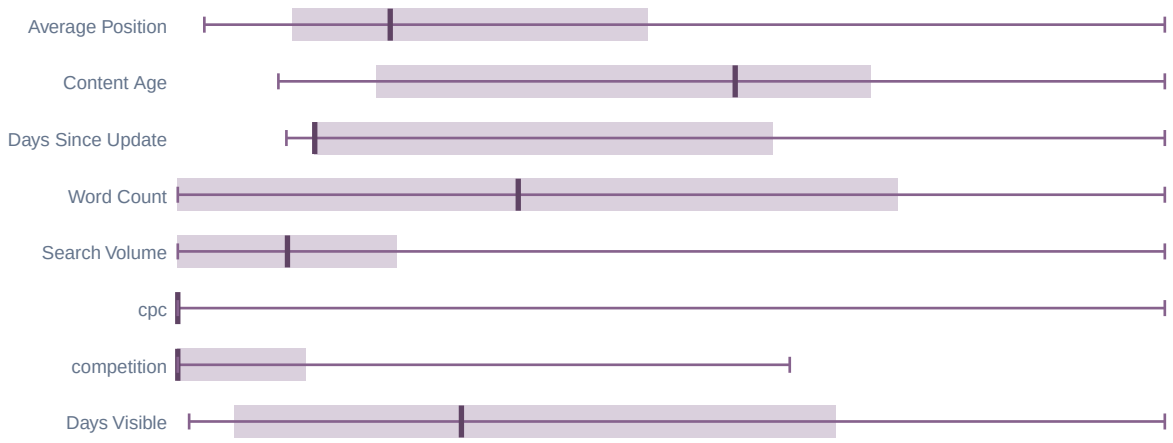


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Chart read: Each row is scaled independently so you can compare shape, not absolute size. Use the table below for the actual values and the median line to see what a typical page looks like for that specific metric.

METRIC	P10	P25	MEDIAN	P75	P90	MEAN
Average Position	1	4.3	8	17.69	37.17	14.43
Content Age	37	73	205	255	363	189.88
Days Since Update	23	29	29	126	209	75.37
Word Count	0	0	1358	2873	3937	1650.6
Search Volume	0	0	10	20	90	131.21
cpc	0	0	0	0	1.57	0.71
competition	0	0	0	0.13	0.62	0.14
Days Visible	1	5	25	58	87	33.51

What To Do With Percentiles

Below P25 means the page is in the bottom quarter for that specific metric. P50 is the typical page. Above P75 means the page is unusually strong on that metric relative to the portfolio, not that the whole portfolio is top tier.

Use these percentiles to see where a page falls relative to the rest of the portfolio. The chart is normalized for readability, so always use the table for real values.

Priority Actions.

8 things you can do right now, ranked by how strong the signal was in our data. Start at #1.

1. **Refresh mature pages before they decay**
   Old pages that get updated bounce back dramatically — this was the single strongest signal in the dataset.
2. **Improve page-one click capture instead of rebuilding from scratch**
   A page already on page 1 returns clicks faster through snippet and clarity improvements than a brand new page can.
3. **Target lower-competition commercial/transactional topics**
   Buy-ready topics with low competition outperform crowded configurations on every metric.
4. **Add real depth to thin pages that already get traffic**
   Adding depth works when it covers more questions. Padding a page with filler to hit a word count doesn't.
5. **Distribute proven pages across more than one channel**
   Strong pages rarely rely on search alone. Support them with email, social, and internal links after publication.
6. **Treat flagged pages as your best optimization targets**
   A page with a diagnosed issue is more actionable than an invisible page with no measurable demand.
7. **Monitor AI traffic separately — it's a different channel**
   AI-attracting pages behave differently from classic organic winners and need their own tracking.
8. **Resolve cannibalization where pages split the same demand**
   76.7M impressions are tied up in overlapping pages. Consolidation and clearer targeting can release that value.

The Content Refresh Playbook.

A step-by-step process for updating content, backed by the freshness and age data in this study.

1. Identify refresh candidates

Why: Pages lose health from 37.2 at 61-90 days to 29.6 by 271-365 days — catch them before the drop

How: List older pages that used to earn impressions or still have meaningful visibility. Start with pages that are clearly dated, incomplete, or no longer competitive.

Expected: Targets the highest-value content most likely to recover

Measure: Track refreshed-page count, health score lift, and 30-day impression recovery versus untreated candidates.

2. Audit and expand thin sections

Why: Top 10% content averages 1.5K words versus 1.3K in the bottom 10%

How: Read the page end to end, identify weak sections, then add missing definitions, examples, comparisons, and current evidence instead of padding.

Expected: Improves depth where missing coverage is holding back an already-visible page

Measure: Track word-count expansion, health score change, and impression lift by refresh batch.

3. Update all statistics, dates, and references

Why: The strongest measured freshness window is 31-90 days at 5.43:1 growth to decline

How: Replace stale figures, update named sources, add recent examples, fix broken links, and make the page visibly current for both users and search engines.

Expected: Aligns the page with the strongest measured freshness band in this dataset

Measure: Track time-to-reindex, 30-day impression lift, and CTR lift after the update goes live.

4. Improve engagement elements

Why: High scroll + high engagement outperforms the weakest bucket by 16.1 health points

How: Add table-of-contents links where useful, break dense sections into scannable blocks, and use visuals or callouts where they actually help comprehension.

Expected: Supports the engagement patterns associated with stronger visibility

Measure: Track scroll depth, engagement rate, and health score before and after layout changes.

5. Re-submit and monitor

Why: Updated content still needs re-crawl and a monitored follow-up window

How: Request recrawl where appropriate, republish the update through your normal distribution channels, and check the page again at 30, 60, and 90 days.

Expected: Keeps the refresh program tied to measured follow-up windows instead of one-time edits

Measure: Track reindex confirmations, ranking recovery, and 30/60/90-day health score trend.

AI Traffic Optimization Guide.

AI traffic (from ChatGPT, Gemini, etc.) is real but small. Here's what we know about it — and what to do.

Pages that attract AI visits have a distinct profile: higher impressions, weaker Google positions, and broader visibility than pages with no AI referrals. Reality check: AI referrals are still only 1.06% of tracked sessions — growing, but small.

What Our Data Shows About AI-Attracting Content

High-AI content in our portfolio averages a weaker Google position but much higher impressions than no-AI content. Across the full available page-history export, the highest AI page rates appear in commercial intent (3.01% of pages with AI referrals) and the 5K+ word-count bucket (24.55% of pages with AI referrals).

Practical Optimization Steps

1. Monitor AI referrals separately from organic clicks and pageview totals.
2. Prioritize clear factual statements, named evidence, and easy-to-scan structure.
3. Review provider mix regularly: OpenAI 4.3K | Gemini 2.3K | Perplexity 823 | Copilot 113 | Claude 87.
4. Start with pages that already get impressions and answer clear questions well.
5. Treat monetization claims cautiously unless attribution is session-level and clean.

Important: we can't yet reliably tie AI traffic to revenue at the individual visit level, so we focus on visibility trends instead of making money claims.

The Quick Wins Checklist.

Actions you can take this week, this month, and on an ongoing basis — ranked by urgency.

TODAY

Fix diagnosed issues on pages that already get traffic

A page with a specific problem you can fix is your fastest win — the upside is measurable and immediate.

TODAY

Review your best 20 striking-distance pages

Pages ranked 11-20 are close to page 1. Small improvements here produce outsized click gains.

THIS WEEK

Refresh your oldest high-visibility pages

Old pages with existing traffic are the best refresh candidates — they've already proven demand.

THIS WEEK

Improve readability on top-performing long pages

Better navigation, clearer sections, and easier scanning support the engagement patterns that stronger pages share.

THIS MONTH

Consolidate overlapping pages that compete with each other

Pick the strongest page, merge the overlapping content, and stop your own URLs from competing against each other.

THIS MONTH

Expand high-potential thin pages with real depth

Focus on thin pages that already get impressions — add missing subtopics and examples instead of bulk-expanding everything.

ONGOING

Keep a rolling refresh calendar

The biggest freshness gains happen when you update content before it fully decays — not after.

ONGOING

Support strong pages with distribution beyond search

Email, social, internal links, and referral placement turn one-channel pages into durable assets.

We Do One Thing: AI SEO.

Everything in this paper runs automatically for every FlyRank client. Here is (most probably not) full list.

We Create Your Content

- Full SEO articles, written and formatted. Not outlines. Not drafts. Done.
- In any language you need. Localized so it reads natural in every market.
- As many as you need. Every day if that's the plan.
- Images, meta titles, and meta descriptions — all generated automatically.
- Optimized for Google and for AI answer engines (ChatGPT, Gemini, Perplexity).

We Publish It

- Straight to your site. Shopify, WordPress, Webflow, Ghost — whatever you use.
- No uploads. No formatting. No copy-paste. It goes live when it's ready.
- Every new page gets submitted to Google for indexing the same day.
- We track whether Google actually picked it up.

We Watch Every Page, Every Day

- Search Console, Analytics, and AI referrals — all in one place.
- Every page gets a health score: impressions, position, click-through rate, and scroll depth.
- We track which pages grow, which flatten, and which fall.
- AI traffic from ChatGPT, Gemini, Perplexity, Copilot, Claude, and Meta — by provider, by day.
- We track when your content peaks, when it starts losing ground, and what queries it actually ranks for.

We Fix What's Not Working.

We don't hand you a report and say good luck. We do the work.

We Tell You What To Fix — And In What Order

- A live optimization queue that updates itself. Not a report you read once.
- Pages that need a refresh, ranked by how much they can recover.
- Pages that get seen but don't get clicked — with the impressions sitting on the table.
- Pages almost on page one, showing what a small push could unlock.
- Quick wins you can act on this week.
- Pages fighting each other for the same keyword — sized by impressions and clicks lost.

We Fix It For You

- Our AI agents rewrite weak titles and descriptions.
- They improve readability and formatting.
- They refresh stale sections in old content.
- They push changes live. Not recommendations. Actual fixes, shipped.

We Keep The Loop Running

- New content gets created and published on schedule.
- Old content gets flagged when it starts to slip.
- Fixes get queued, made, and tracked.
- This runs every week. Not a one-time audit.

The Guarantee

3x your reach in 3 months. Or your money back.

What We Optimize.

This appendix is additive. The existing report above stays intact; the pages below define a new local-only optimization framework for content creation and refresh.

212.4K
pages scored

90.6K
pages with query coverage

42.7%
query-covered share

CREATION INPUTS

word_count
char_count
content_type
main_intent
provider_used
model_used
publish_month
publish_quarter

REFRESH INPUTS

days_since_update
freshness_tier
content_age_days
age_freshness_interaction
age_freshness_numeric
update_recency_bucket

CONTEXT / CONFOUNDERS

client_id
search_volume
cpc
competition
competition_level

EXCLUDED / LEAKY

health_score
health_label
needs_indexing
is_quick_win
needs_ctr_fix
needs_engagement_fix
• +4 more

Primary Outputs
core_quality_score, relevance_adjusted_quality_score, opportunity_score, trend_up_vs_down

Secondary Diagnostics
impressions_90d, clicks_90d, sessions_90d, ctr, engagement_rate, scroll_rate, ai_sessions_90d, ai_traffic_pct, avg_position

Significance Policy
Stable signals must clear sample-size gates, FDR correction, group-aware model support, and bootstrap stability. If a subsection has no stable rows, the appendix falls back to directional rows instead of leaving the page blank.

Why The New Score Exists.

The legacy score stays in the report above. This section adds a second framework that reduces raw-position fragility and discounts off-target visibility when local query history exists.

Why Not Let Raw Position Dominate?

A page can get noisy early impressions or surface for mismatched demand. That can distort average position without proving the content is actually good. The new score relies more on demand capture, consistency, engagement, and relevance correction.

How Relevance Correction Works

Pages with query-history coverage get a multiplier based on exact-match share, close-match share, off-target share, and the bucket of the top-ranking query. Pages without query coverage keep the core score and receive a lower confidence label instead of a fabricated relevance adjustment.

CORE QUALITY	RELEVANCE-ADJUSTED	OPPORTUNITY
MEDIAN 45.6	MEDIAN 40.1	MEDIAN 49.5
P90 74.0	P90 64.1	P90 57.8
MEAN 50.0	MEAN 43.1	MEAN 48.4

Legacy vs New

HEALTH -> CORE	HEALTH -> RELEVANCE
0.729	0.591
TOP DECILE OVERLAP	BOTTOM DECILE OVERLAP
36.9%	16.8%

Disagreement Slice

COHORT SIZE	QUERY-COVERED
21.2K	6.9%
AVG OFF-TARGET SHARE	
1.8%	

Chart read: High overlap means the old and new systems often agree. The disagreement cohort matters because it isolates pages where raw visibility and relevance-corrected quality tell different stories.

Most Significant Creation Inputs.

Stable creation-side signals are preferred. When none clear the full threshold for a target, this page falls back to the strongest directional rows so the appendix still shows what the local data is pointing toward.

TOP CREATION SIGNALS

TARGET	INPUT	EFFECT	95% CI	STABLE	N	METH
Core Quality	**Provider Used**	28.19	15.59 / 31.16	69%	212.4K	2
Core Quality	**Model Used**	15.46	9.27 / 26.20	69%	212.4K	2
Core Quality	**Main Intent**	15.30	8.20 / 20.40	78%	212.4K	2
Core Quality	**Competition Level**	13.47	5.38 / 18.97	94%	212.4K	2

How To Read This Table

Effect shows the signed univariate relationship or strongest level contrast. Stability is the share of client-group bootstraps that preserved the direction. Meth counts how many supporting methods cleared the threshold.

- High method support = stronger evidence
- Creation inputs are publish-time decisions

Most Significant Refresh Inputs.

Refresh-side rows can represent stable quality-recovery signals or strong directional opportunity signals. The important distinction is that these are post-publication levers, not publish-time choices.

TOP REFRESH SIGNALS

TARGET	INPUT	EFFECT	95% CI	STABLE	N	METH
Opportunity	**Age x Freshness (Numeric)**	0.48	0.70 / 0.90	100%	212.4K	4
Opportunity	**Content Age**	0.41	0.66 / 0.90	100%	212.4K	4
Opportunity	**Days Since Update**	0.40	0.61 / 0.87	100%	212.4K	4
Opportunity	**Character Count**	-0.28	-0.69 / -0.20	100%	212.4K	4

Refresh vs Opportunity

Quality and opportunity are split on purpose. A page can be low quality and low opportunity, or mediocre quality with strong upside. The refresh table surfaces both kinds of signal without collapsing them into one label.

Legacy vs New: Where They Diverge.

The legacy system remains above. This closing page shows where the new framework agrees with it, where it meaningfully diverges, and which growth-side inputs still matter after the new scoring logic is applied.

LEGACY SNAPSHOT

ROWS ANALYZED
211.1K

RF R^2 TEST
0.995

GROWTH ACCURACY
0.679

SCORE AGREEMENT

TOP OVERLAP
36.9%

BOTTOM OVERLAP
16.8%

DISAGREEMENT COHORT
21.2K

RISK PATTERN

QUERY-COVERED
6.9%

AVG OFF-TARGET SHARE
1.8%

TOP GROWTH SIGNALS

INPUT	EFFECT	95% CI	STABLE	N	METH
Content Age	-0.21	-0.57 / -0.02	99%	120.7K	3

CONTENT ARCHETYPES UNDER THE NEW SYSTEM

Archetype 1 (2.3% of sampled pages)

Quality 60.8 | Opportunity 45.6 | Confidence 52.2 | Mode intent informational

Archetype 2 (15.9% of sampled pages)

Quality 57.4 | Opportunity 43.8 | Confidence 68.0 | Mode intent informational

Archetype 3 (31.9% of sampled pages)

Quality 49.4 | Opportunity 51.2 | Confidence 69.7 | Mode intent informational

Chart read: The disagreement slice is the practical migration queue. Those are the pages most likely to be overrated by raw visibility or underrated by the old system's structure.

Which Pages Will Grow?

A gradient-boosted classifier trained on 120.7K pages with clear up/down trends. Validated with GroupKFold (no client leakage).

AUC (5-fold)	Accuracy	True Positives
0.807	72.6%	62.0K
± 0.075	± 12.8%	of 74.5K growing

WHAT DRIVES GROWTH (PERMUTATION IMPORTANCE ON AUC)

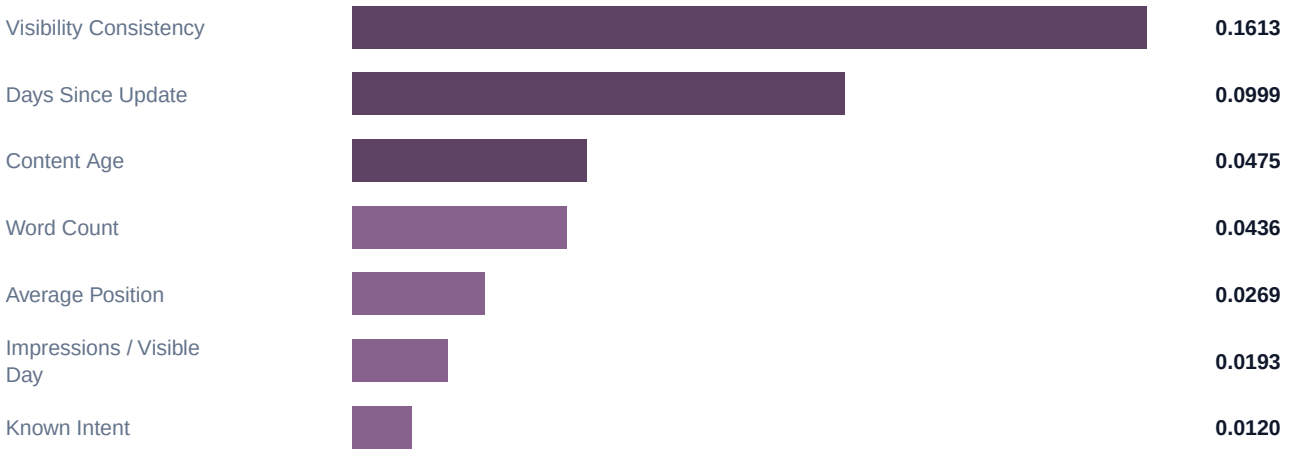


Chart read: Permutation importance measures how much AUC drops when a feature is shuffled. Higher = more important. The top 3 drivers are visibility consistency, freshness, and content age.

MODEL CALIBRATION

When the model says a page has an 80% chance of growing, do 80% actually grow?

Predicted	Actual	n
4.0%	4.7%	n=5.5K
15.3%	23.0%	n=3.6K
24.3%	37.5%	n=8.7K
35.2%	37.0%	n=7.2K
45.0%	41.4%	n=12.8K
54.6%	47.6%	n=13.3K
64.7%	54.1%	n=15.6K
74.4%	71.3%	n=13.9K
83.9%	87.7%	n=14.8K
95.8%	97.2%	n=25.3K

Chart read: Green means predicted and actual rates are close. The model is well-calibrated at the extremes (very likely / very unlikely to grow) and slightly optimistic in the middle range.

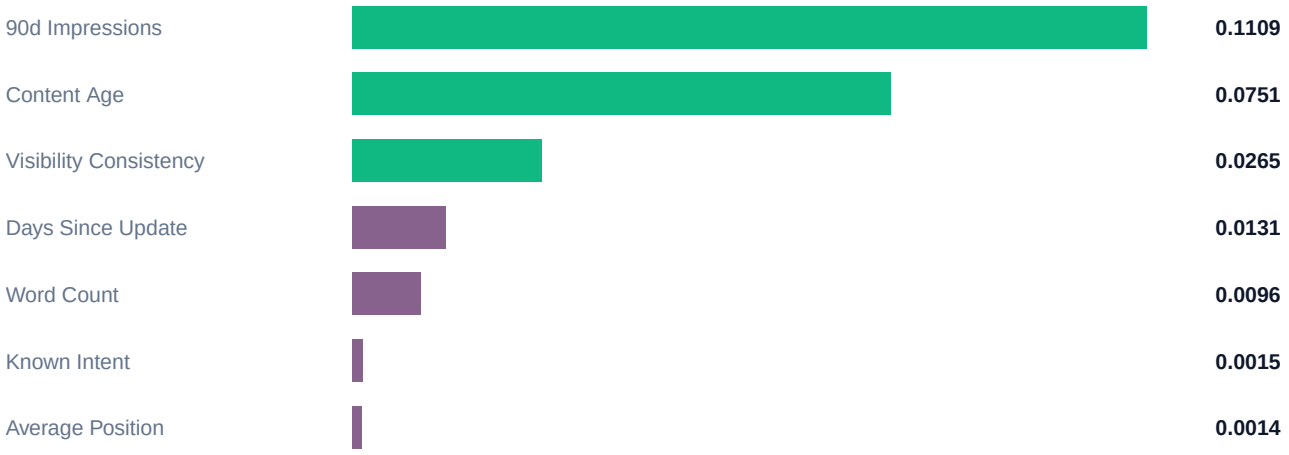
Visibility consistency (days with impressions) is the #1 growth driver. Pages visible 31+ days in the 90-day window have the highest growth probability. Freshness and age are the next biggest levers — younger, recently updated pages grow more.

Which Zombie Pages Can Recover?

65.7K pages had zero impressions and clicks in the previous 30 days. 59.2% recovered naturally. Can we predict which ones?

AUC	Accuracy	Recovery Rate
0.943	87.4%	59.2%
± 0.056		38.9K of 65.7K

WHAT PREDICTS RECOVERY (PERMUTATION IMPORTANCE)



Recoverable Zombie Profile

Median age: 37 days
Median words: 2.7K
Known intent: 87%
Low competition: 72%
Actual recovery: 92%

Unrecoverable Zombie Profile

Median age: 238 days
Median words: 1.6K
Known intent: 74%
Low competition: 56%
Actual recovery: 2%

Recoverable zombies are young (37 days), have known intent, and had some impressions in the 90-day window. Unrecoverable zombies are old (238 days) with zero 90-day impressions. Age and prior impressions are the two strongest signals — a page that has never shown up in 90 days of data is almost certainly dead.

Predicting 30-Day Momentum.

For 112.0K pages with meaningful traffic, can we predict which will improve > 10% next month?

AUC	Accuracy	Improved Rate
0.915	83.0%	56.1%
± 0.046		base rate

WHAT PREDICTS NEXT-MONTH IMPROVEMENT

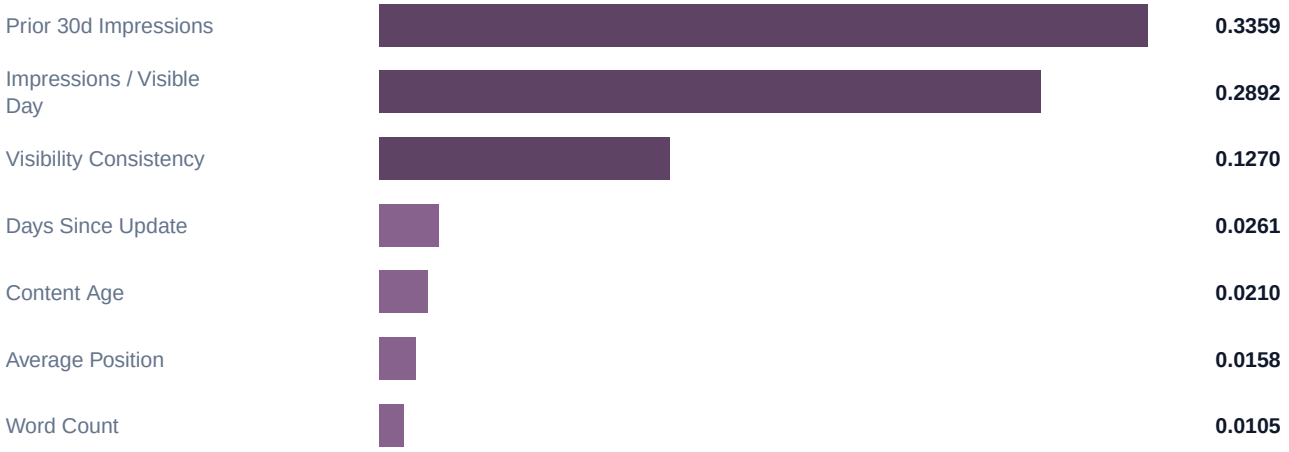


Chart read: Current impression volume and impressions-per-visible-day are the strongest momentum signals. Pages that are already earning visibility efficiently are most likely to improve further.

This model identifies pages on an upward trajectory. Use it to prioritize which pages to support (add internal links, expand content, improve metadata) while they still have momentum. The strongest signal is not content quality — it's current visibility efficiency.

Refresh Impact: Quantified.

Mann-Whitney U tests comparing recently refreshed (d30d) vs stale pages, stratified by age and competition. All 0 of 0 strata are significant at p < 0.001.

MEDIAN IMPRESSION LIFT BY AGE × COMPETITION (RECENTLY REFRESHED VS STALE)

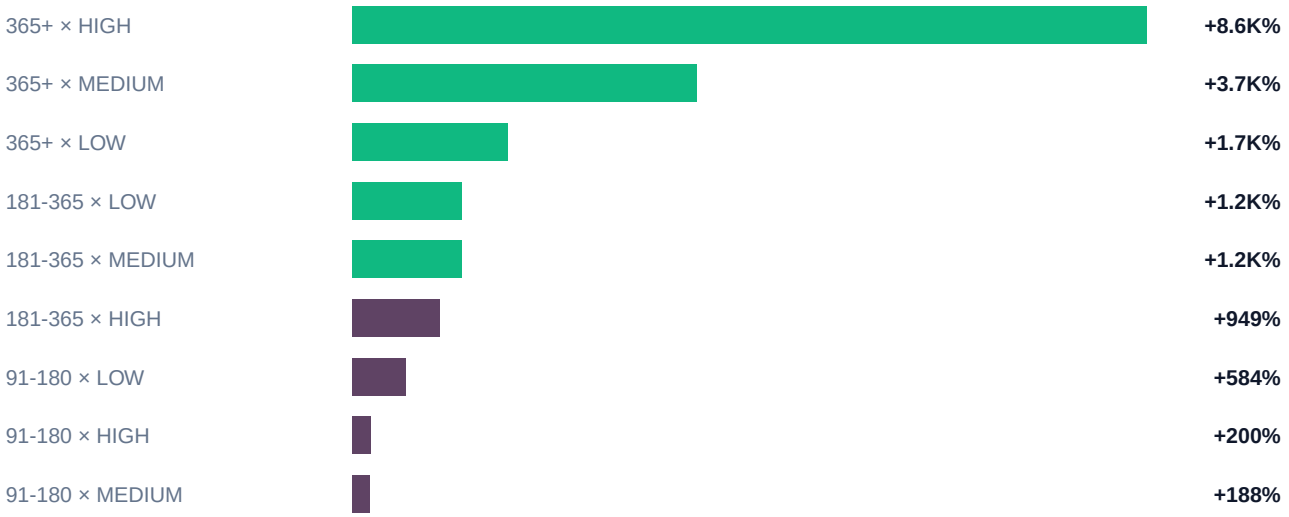


Chart read: Every age × competition combination benefits from refreshing. Older pages get the most dramatic lifts — 365+ day pages see 1,694% to 8,638% median impression increases after a refresh.

FLOWCHART DECISION EFFECT SIZES (BOOTSTRAP 95% CI)

Each row quantifies the median impression difference for a key flowchart decision, with 2,000 bootstrap samples for confidence intervals.

Competition = LOW	LOW vs HIGH competition	**+125**	[91, 164]
Intent clear	Known intent vs unknown/unclassified	**+289**	[261, 318]
Word count 5K+	5K+ words vs 2K-3.5K words	**+3.4K**	[3.2K, 3.7K]
Refresh old pages	Refreshed (<=30d) vs stale (>90d) for 180+ day pages	**+600**	[551, 649]

All refresh comparisons use Mann-Whitney U (non-parametric, no normality assumption) with bootstrap 95% confidence intervals. Effect sizes are median differences, not means, to resist outlier distortion. Every stratum shown has p < 0.001.

Where the Numbers Jump.

Fine-grained binning reveals non-linear thresholds where small input changes create large outcome shifts.

WORD COUNT ' MEDIAN IMPRESSIONS (ACTIVE PAGES)

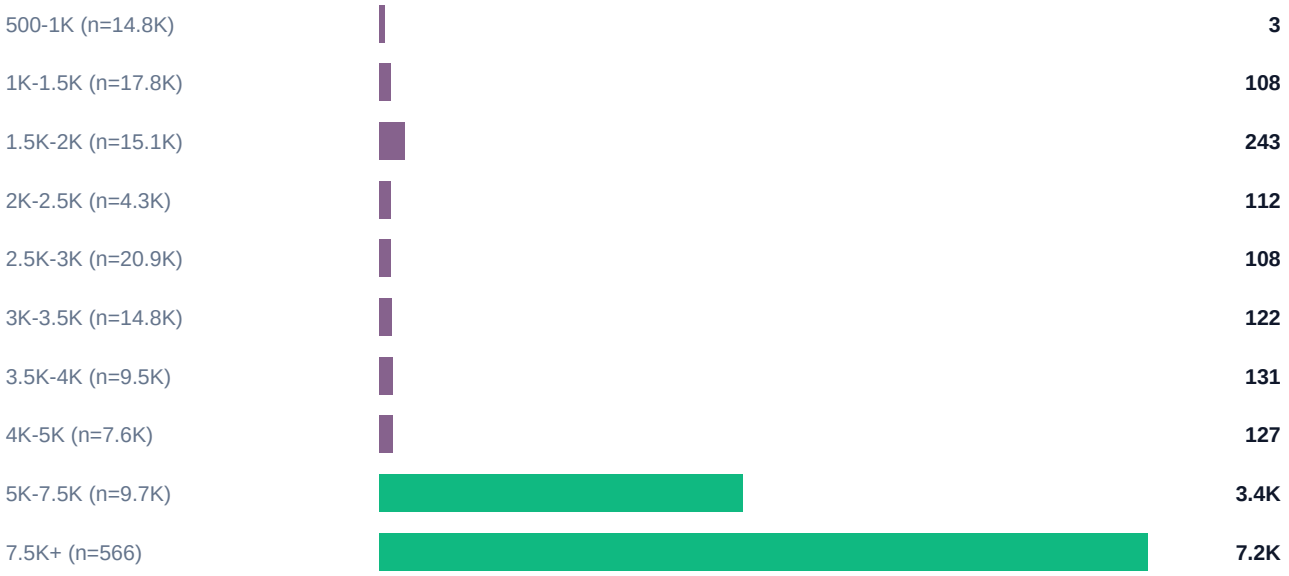


Chart read: The jump from 4K-5K words (median 131 impressions) to 5K-7.5K (median 3,421) is 26x. Below 5K, word count barely moves impressions. Above 5K, it creates a step change — but only for topics that genuinely warrant the depth.

VISIBILITY CONSISTENCY ' MEDIAN IMPRESSIONS

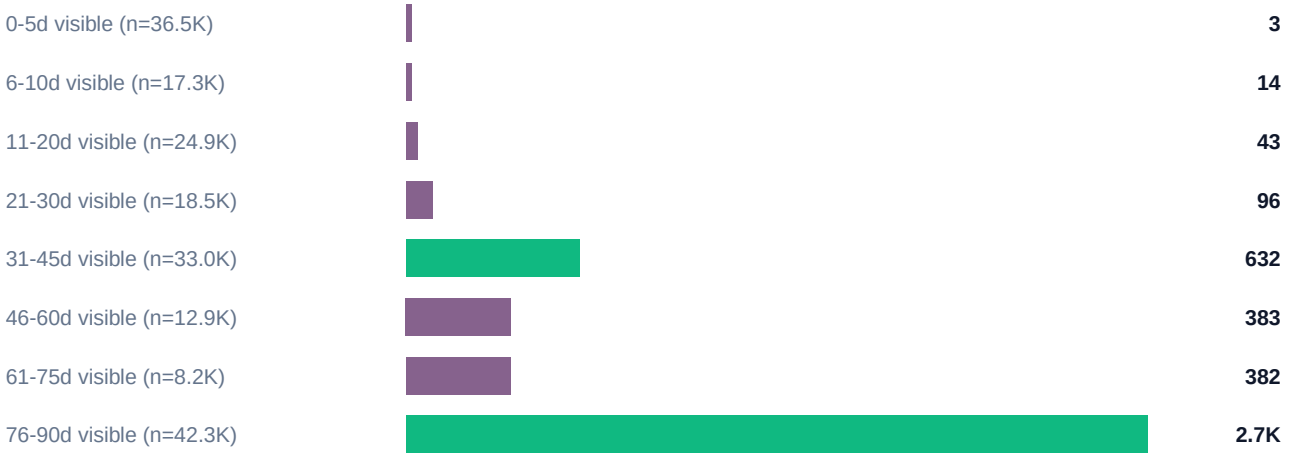


Chart read: Pages visible 31-45 days hit a growth inflection: median impressions jump to 632 and growth rate peaks at 86%. Pages visible 76-90 days reach 2,700 median impressions but growth rate drops to 33% — they've plateaued.

Linear correlations missed these effects because the relationship between inputs and outputs is not a straight line. Word count has near-zero linear correlation with impressions ($r = 0.02$), but the 5K+ bin has 26x the median impressions of the 4K-5K bin. Always check for step-changes, not just correlations.

Every Flowchart Decision Tested.

Kruskal-Wallis H tests and Mann-Whitney U tests confirm that every major grouping in the playbook produces statistically significant differences in impressions.

Test	Method	Statistic	p-value
Competition Level ' Impressions	Kruskal-Wallis H	913	< 0.001 Sig.
Intent ' Impressions	Kruskal-Wallis H	1.2K	< 0.001 Sig.
Freshness Tier ' Impressions	Kruskal-Wallis H	33.9K	< 0.001 Sig.
Recently Refreshed (<=30d) vs Stale (>90d) ' Impressions	Mann-Whitney U	4.3B	< 0.001 Sig.
Word Count Tier ' Impressions	Kruskal-Wallis H	26.3K	< 0.001 Sig.
Position Tier ' CTR	Kruskal-Wallis H	10.2K	< 0.001 Sig.

WHAT MODELS WORKED VS FAILED

Model	AUC / R2	Verdict
Momentum Prediction	0.915 AUC	Strong. Predicts 30-day trajectory.
Zombie Recovery	0.943 AUC	Excellent. Separates recoverable from dead.
Growth Prediction	0.807 AUC	Good. Identifies likely growers.
CTR from Content	Negative R2	Failed. CTR depends on metadata, not content.
Impressions from Inputs	Negative R2	Failed. Can't predict from creation-time inputs.

All predictive models use GroupKFold cross-validation grouped by client_id — no client appears in both train and test. Feature importance is measured by permutation importance (shuffling features and measuring AUC drop), not tree-based importance which overstates correlated features. Effect sizes use bootstrap confidence intervals with 2,000 resamples.